TITLE : CRSM

(Computer Repair Shop Management System)

PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE

S.I.E.S College of Arts, Science and Commerce Sion(W), Mumbai-400 022

"COMPUTER REPAIR SHOP MANAGEMENT SYSTEM " (CRSM)

For the Partial Fulfilment for The Degree of Bachelor of Science (Computer Science) 2024-2025

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CERTIFICATE

This is to certify that **MR. HARIHARAN RAJU** Roll No. **TCS2425018** has Successfully completed the project work entitled **COMPUTER REPAIR SHOP MANAGEMENT SYSTEM (CRSM)** as the partial fulfilment of B.Sc. (Computer Science)during the course of **TYBSc Computer Science [Semester- 6]**

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INDEX

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INTRODUCTION

In today's digital world, computers play a crucial role in both personal and professional settings. As technology continues to advance, the need for efficient and reliable computer repair services has significantly increased. However, many repair shops still rely on manual record-keeping, leading to inefficiencies, mismanagement of repair jobs, inventory shortages, and poor customer service. Managing repairs, tracking inventory, handling transactions, and maintaining customer relationships can be challenging without a proper system in place. To address these issues, the Computer Repair Shop Management System (CRSMS) has been developed as a comprehensive web-based solution to streamline and automate the workflow of computer repair businesses.

The system is designed to replace traditional manual processes with an automated platform that efficiently manages repair jobs, inventory tracking, customer interactions, and financial transactions. It provides a centralized database where all essential information is securely stored and easily accessible. Built using **PHP, MySQL, JavaScript, Bootstrap, and AdminLTE**, the system ensures a user-friendly experience with an intuitive interface that can be operated by both technical and non-technical users. The CRSMS assigns different roles such as **Administrator, Staff, and Technician**, ensuring that each user has specific access privileges based on their responsibilities. This role-based access control enhances security and accountability within the organization.

One of the key objectives of this system is to simplify and improve the efficiency of repair job tracking. Technicians can update the status of ongoing jobs in real time, allowing shop owners and customers to stay informed. Additionally, the system provides an automated inventory management module that keeps track of stock levels, preventing shortages or overstocking of computer parts and accessories. By integrating financial management features, the system also ensures accurate record-keeping of all transactions, reducing human errors and enhancing business transparency.

Furthermore, the system enhances customer experience by providing timely updates on repair status and maintaining a service history for personalized interactions. Customers no longer need to visit the shop or make phone calls to inquire about their repair progress, as automated notifications keep them informed. The system

also includes reporting and analytics tools, which allow shop owners to monitor business performance, analyze sales trends, and make data-driven decisions.

Overall, the **Computer Repair Shop Management System** is a modern, scalable, and cost-effective solution that aims to improve workflow efficiency, enhance service quality, and increase customer satisfaction. By adopting this system, repair shops can optimize their daily operations, reduce human errors, and focus on delivering high-quality technical support to their clients.

2.1 System Description

The **Computer Repair Shop Management System (CRSMS)** is a web-based application developed to automate and enhance the management of repair shops by replacing inefficient manual processes. The system integrates various functionalities, including repair job tracking, inventory management, financial transaction handling, and customer relationship management, ensuring seamless business operations.

• **User Roles and Access Control:** The system provides a role-based access mechanism, where users are categorized as **Administrator, Staff, and Technician**. Administrators have full system control, allowing them to manage services, products, and users. Staff members can handle customer interactions, record transactions, and update stock levels, while technicians focus on managing and updating assigned repair jobs.

• **Repair Job Management:** The system simplifies the process of logging repair requests, assigning tasks to technicians, and tracking progress in real-time. Automated notifications keep customers informed about their repair status, reducing manual follow-ups and improving service efficiency.

• **Inventory Management:** The system includes real-time inventory tracking to monitor stock levels of computer parts and accessories. It provides **low-stock alerts** to prevent shortages, ensuring that necessary parts are always available for repairs and sales.

• **Transaction and Financial Management:** The system securely records all **sales and service payments**, generates **invoices and receipts**, and maintains a **detailed financial history**. This minimizes manual bookkeeping errors and simplifies financial tracking.

• **Customer Relationship Management (CRM):** The system stores customer details and repair histories, enabling personalized service. It allows repair shops to track past transactions and interactions, improving customer communication and satisfaction.

• **Reporting and Business Analytics:** The system generates **real-time reports** on **sales trends, repair job completion, and inventory movement**. These insights help shop owners make **data-driven decisions** to improve operational efficiency and profitability.

• **Security and Data Management:** The system ensures **secure storage** of sensitive customer and business data through **role-based authentication, database encryption, and regular backups**. This prevents unauthorized access and data loss.

• **User-Friendly Interface:** Developed with **PHP, MySQL, JavaScript, Bootstrap, and AdminLTE**, the system features an **intuitive and responsive design** that is easy to navigate, ensuring accessibility across multiple devices.

2.2 Objective

The **Computer Repair Shop Management System (CRSMS)** is designed to address the operational challenges faced by repair shops by integrating an automated platform for managing repair jobs, inventory, financial transactions, and customer interactions. The system enhances efficiency, reduces errors, and improves customer satisfaction.

• **Streamlining Repair Job Management:** The system provides a centralized platform to track and manage all repair requests. Users can log new repair jobs, assign tasks to technicians, and update job statuses in real-time, ensuring efficient and timely service completion.

• **Improving Inventory Management:** The system enables real-time inventory tracking to monitor available parts and products, reducing the risk of stock shortages or overstocking. Automated alerts help maintain necessary stock levels for uninterrupted repair services.

• **Enhancing Financial Tracking:** Integrated transaction management ensures accurate tracking of sales, purchases, and service payments. The system automates invoice generation and financial reporting, simplifying bookkeeping and improving accuracy.

• **Improving Customer Communication:** The system provides automated real-time updates on repair status, reducing the need for manual follow-ups. Customers receive timely notifications, enhancing their experience and satisfaction.

• **User Access Management:** The system implements role-based user management, ensuring only authorized personnel can access specific modules. This enhances security while enabling better task delegation and workflow management.

• **Facilitating Data-Driven Decision-Making:** The system provides comprehensive reports and analytics on sales trends, repair turnaround times, and inventory usage. This data helps shop owners make informed decisions to improve profitability and efficiency.

• **Promoting Efficient Resource Allocation:** The system allows shop owners to effectively manage technician workloads by tracking assigned and completed jobs. This ensures balanced workload distribution and prevents overburdening employees.

• **Supporting Customer Relationship Management (CRM):** Customer details and repair history are stored within the system, allowing for personalized interactions. This improves customer retention and helps repair shops build long-term relationships with clients.

• **Ensuring Compliance with Standards:** The system includes documentation and reporting tools that help repair shops maintain compliance with industry regulations, particularly regarding data security and financial transparency.

• **Enhancing User Experience:** The system is designed with an intuitive and user-friendly interface, ensuring easy navigation and minimal learning curve for employees. This allows for quick adoption and effective utilization of system features.

2.3 Proposed System

The **Computer Repair Shop Management System (CRSMS)** is designed to replace traditional manual processes with an automated web-based solution that improves business efficiency and customer service. The system integrates various modules to enhance workflow and management capabilities.

• **Centralized Database Management:** The system provides a unified platform to store customer details, repair job records, inventory data, and financial transactions. This ensures quick and secure access to business-critical information.

• **Automated Repair Job Tracking:** The system allows users to log repair jobs, assign them to technicians, and update job status in real-time. Customers receive automated notifications about their service progress.

• **Efficient Inventory Management:** The system maintains real-time stock levels and sends alerts when inventory is low, ensuring timely restocking of essential components and preventing delays in repairs.

• **Comprehensive Transaction Management:** The system enables the secure recording of sales, service payments, and refunds. It generates invoices, receipts, and detailed transaction history for accurate financial tracking.

• **Enhanced Customer Relationship Management (CRM):** Customer details, service history, and transaction records are stored within the system, enabling personalized interactions and improved service management.

• **Detailed Reporting and Analytics:** The system generates reports on repair job statuses, inventory trends, and sales performance. These insights assist business owners in making data-driven decisions.

• **Role-Based Access Control:** The system categorizes users into **Administrator, Staff, and Technician**, restricting access to authorized functionalities based on roles to ensure security and efficient workflow management.

• **Secure and Scalable Architecture:** The system includes security measures such as authentication, encryption, and regular backups, ensuring data protection. It is designed to be scalable, allowing for future expansion as the business grows.

2.4 Stakeholders

The **Computer Repair Shop Management System (CRSMS)** is designed to serve multiple stakeholders involved in the repair shop's daily operations, ensuring efficient management and seamless coordination among users.

• **Repair Shop Owner/Administrator:** Responsible for overseeing shop operations, managing system users, and analyzing reports for business performance and growth. The administrator has full access to all system features.

• **Staff Members:** Handle customer interactions, log repair jobs, process transactions, and update inventory records. Staff members play a crucial role in ensuring smooth day-to-day operations.

• **Technicians:** Assigned to repair jobs and responsible for updating job progress in real-time. They ensure timely and accurate repair service delivery.

• **Customers:** Submit repair requests, purchase products, and receive updates regarding the status of their repairs. The system enhances customer engagement by providing transparency and efficient communication.

• **Software Developers:** Responsible for system maintenance, updates, and ensuring the security and scalability of the application.

• **Suppliers:** Provide computer parts and accessories required for repairs and inventory management. Integration with suppliers ensures better stock planning and order management.

2.5 Existing System and Its Demerits

Traditional computer repair shops often rely on manual processes for managing repairs, inventory, and transactions, leading to inefficiencies and inaccuracies. The limitations of the existing system include:

• **Inefficient Repair Job Tracking:** Manual logs and paperwork make it difficult to track repair jobs, leading to misplaced records and delays in service completion.

• **Poor Inventory Management:** Without an automated system, tracking stock levels is challenging, resulting in frequent shortages or over-purchasing of repair components.

• **Manual and Error-Prone Transactions:** Paper-based invoicing and financial records increase the risk of errors and miscalculations, leading to financial discrepancies.

• **Lack of Customer Communication:** Customers must visit the shop or make phone calls to check their repair status, which is time-consuming and inconvenient.

• **Data Inaccessibility and Security Risks:** Paper records can be lost, damaged, or accessed by unauthorized personnel, compromising data security and operational efficiency.

• **Time-Consuming Operations:** Employees must manually search for records, slowing down service delivery and reducing overall productivity.

2.6 Advantages of the Proposed System

The **Computer Repair Shop Management System (CRSMS)** addresses the shortcomings of the existing system by automating core processes and enhancing business efficiency.

• **Automated Repair Tracking:** The system provides a centralized platform to track repair jobs, ensuring timely updates and reducing the risk of misplaced records.

• **Real-Time Inventory Management:** Stock levels are automatically updated, and low-stock alerts prevent shortages, ensuring seamless repair operations.

• **Streamlined Financial Transactions:** The system automates invoicing and receipt generation, reducing errors and improving financial accuracy.

• **Improved Customer Service:** Customers receive automated notifications about their repair status, enhancing transparency and engagement.

• **Enhanced Security and Data Accessibility:** The system ensures role-based access control, secure data encryption, and regular backups to protect business information.

• **Increased Productivity and Cost Efficiency:** Automation reduces manual workload, allowing employees to focus on essential tasks and improving overall business profitability.

2.7 Feasibility Study

Before implementing the **Computer Repair Shop Management System (CRSMS)**, a feasibility study is conducted to evaluate its viability in various aspects.

• **Technical Feasibility:** The system is developed using widely used technologies such as **PHP, MySQL, JavaScript, and Bootstrap**, ensuring ease of maintenance and scalability. It is compatible with standard hardware and can be deployed on local or cloud servers.

• **Economic Feasibility:** Automating repair shop operations reduces manual labor costs and improves revenue by enhancing service efficiency. The system's long-term benefits outweigh the initial investment.

• **Operational Feasibility:** The system streamlines operations by automating repair tracking, inventory management, and financial transactions. It is designed with an intuitive interface for easy adoption by employees.

• **Legal Feasibility:** The system ensures secure storage of customer data, compliance with financial reporting standards, and adherence to data protection regulations.

2.8 Requirement Specification

The **Computer Repair Shop Management System (CRSMS)** requires specific **hardware, software, functional, and non-functional** components to ensure smooth operation, security, and user accessibility.

Functional Requirements

• **User Management:** Secure authentication and role-based access for **Administrator, Staff, and Technician** users.
• **Repair Job Management:** Log repair jobs, assign technicians, update statuses, and notify customers.
• **Inventory Management:** Track stock levels and send low-stock alerts.
• **Transaction Management:** Record sales, generate invoices, and maintain transaction history.
• **CRM:** Store customer details and service history for personalized interactions.
• **Reporting & Analytics:** Generate real-time reports on sales, inventory, and repairs.
• **Security & Data Protection:** Implement role-based access, encryption, and backups.
• **User-Friendly Interface:** Ensure **easy navigation and fast response times**.

Non-Functional Requirements

• **Performance:** Support multiple users simultaneously without delays.
• **Scalability:** Allow future expansion and feature integration.
• **Security:** Protect data with encryption, authentication, and access controls.
• **Reliability:** Ensure high availability with automatic backups.

Hardware Requirements

• **Server:** Intel Core i5+, 8GB RAM, 200GB SSD, stable internet.
• **Client:** Dual-core processor, 4GB RAM, internet access.

Software Requirements

- **OS:** Windows 10/11, Linux, macOS (for local deployment).
- **Database:** MySQL/MariaDB.
- **Development:** PHP, JavaScript, HTML, CSS, Bootstrap, AdminLTE.
- **Tools:** XAMPP, Visual Studio Code, GitHub.

MODULES AND UNIT

1. **User Management Module**
    - **Purpose:** To handle user accounts and roles within the system.
    - **Components:**
        - **Add New User:** Allows the Administrator to create new user accounts with specified roles.
        - **List All Users:** Displays all registered users in the system.
        - **View User Details:** Provides detailed information about a selected user.
        - **Edit User Details:** Allows modification of user information.
        - **Delete User Details:** Enables the removal of a user from the system.
        - **Update Account Details/Credentials:** Allows users to change their passwords and personal information.
2. **Service Management Module**
    - **Purpose:** To manage services offered by the repair shop.
    - **Components:**
        - **Add New Service:** Enables the Administrator to add new services to the system.
        - **List All Services:** Displays a list of all services offered by the shop.
        - **View Service Details:** Provides detailed information about a specific service.

- **Delete Service:** Allows the Administrator to remove a service from the list.

3. **Product Management Module**
   - **Purpose:** To manage the inventory of products available in the shop.
   - **Components:**
     - **Add New Product:** Enables the addition of new products to the system.
     - **List All Products:** Displays all available products in the shop.
     - **View Product Details:** Provides detailed information about a selected product.
     - **Update Product Details:** Allows modification of product information.
     - **Delete Product:** Enables the removal of a product from the inventory.
4. **Transaction Management Module**
   - **Purpose:** To manage transactions related to services and products sold.
   - **Components:**
     - **Add New Transaction:** Allows the addition of new transactions for services and products.
     - **List All Transactions:** Displays a list of all transactions conducted.
     - **View Transaction Details:** Provides detailed information about a specific transaction.
     - **Update Transaction Details:** Allows modification of existing transaction records.
     - **Update Transaction Status:** Enables the updating of the status of transactions (e.g., completed, pending).

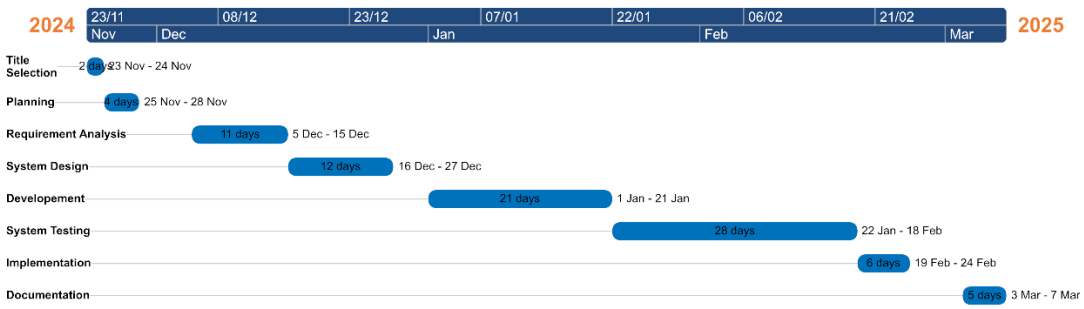
- **Print Transaction Details:** Generates a printable version of the transaction for records.
- **Delete Transaction:** Allows the removal of a transaction from the system.

5. **Public Site Module**
   - **Purpose:** To provide a front-end interface for clients and potential clients.
   - **Components:**
     - **Home Page:** Displays welcome content and key information about the shop.
     - **About Us Content:** Provides background information about the shop.
     - **List All Services Offered:** Displays all the services available to clients.
     - **Search Services:** Enables users to search for specific services.
     - **List All Products Offered:** Displays all products available for sale.
     - **Search Products:** Allows users to search for specific products.
6. **Dashboard Module (for Administrators and Staff)**
   - **Purpose:** To provide an overview of the system's key metrics and quick access to functionalities.
   - **Components:**
     - **Home Page:** Displays a summary of key performance indicators, such as total transactions, number of services, and products.
     - **Visual Reports:** Offers graphical representations of data (e.g., charts showing sales trends).

STATUS CHART (GANTT CHART)

Task	Planned Date / Actual Date	Start Date	End Date	Duration (Days)
Title Selection	Planned Date	23-Nov-24	24-Nov-24	1
	Actual Date	23-Nov-24	24-Nov-24	1
Planning	Planned Date	24-Nov-24	27-Nov-24	3
	Actual Date	24-Nov-24	26-Nov-24	2
Requirement Analysis	Planned Date	05-Dec-24	14-Dec-24	9
	Actual Date	05-Dec-24	15-Dec-24	10
System Design	Planned Date	15-Dec-24	24-Dec-24	9
	Actual Date	15-Dec-24	27-Dec-24	12
Development	Planned Date	01-Jan-25	21-Jan-25	20
	Actual Date	01-Jan-25	24-Jan-25	23
System Testing	Planned Date	22-Jan-25	14-Feb-25	24
	Actual Date	22-Jan-25	18-Feb-25	27
Implementation	Planned Date	16-Feb-25	22-Feb-25	6
	Actual Date	16-Feb-25	24-Feb-25	8
Documentation	Planned Date	02-Mar-25	04-Mar-25	2
	Actual Date	03-Mar-25	07-Mar-25	4

CRSM Status Chart



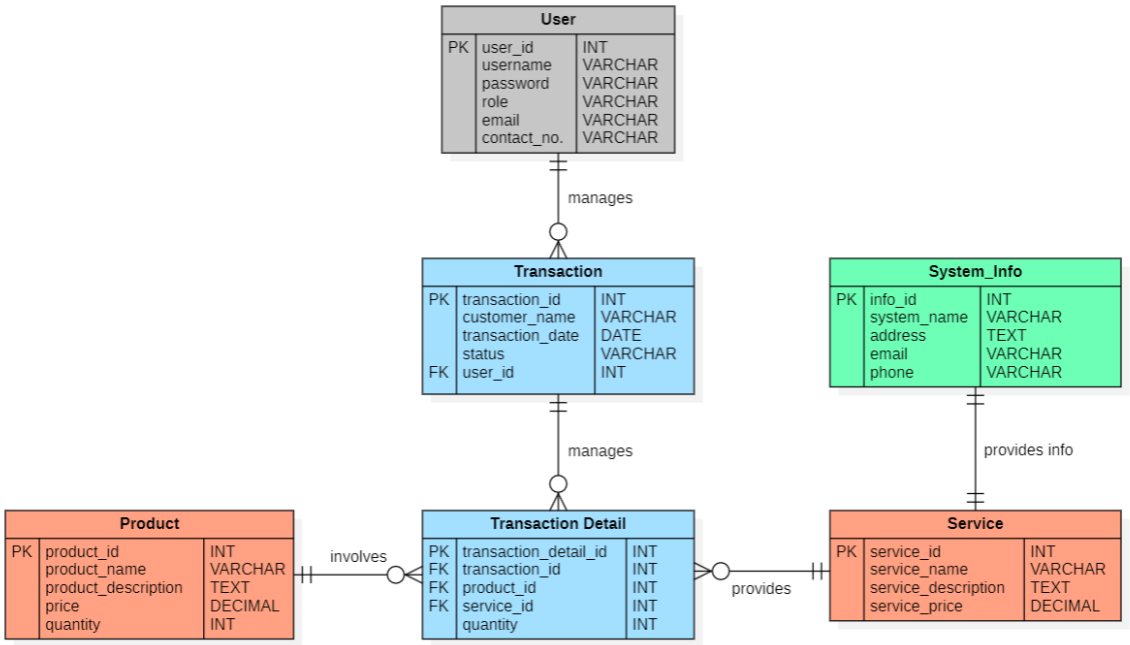
EVENT TABLE

Event Number	Event Description	Trigger/Condition	Actions/Processes
1	User Login	User logs into the system	Verify credentials, grant access
2	New Repair Request	Customer submits a repair request	Log repair details, assign technician
3	Inventory Update	Technician updates part usage	Deduct used parts, notify stock management
4	Payment Processing	Customer makes payment	Process payment, generate invoice
5	Repair Completion	Technician marks job as complete	Notify customer, update job status
6	Order Parts	System detects low stock	Place order for new parts, update inventory
7	Customer Inquiry	Customer inquires about repair status	Retrieve and display repair progress
8	Technician Assignment	New repair job added	Assign technician, notify staff
9	System Backup	Scheduled system maintenance	Backup database, store logs
10	Error Handling	System encounters an issue	Log error, notify administrator

SYSTEM DESIGN

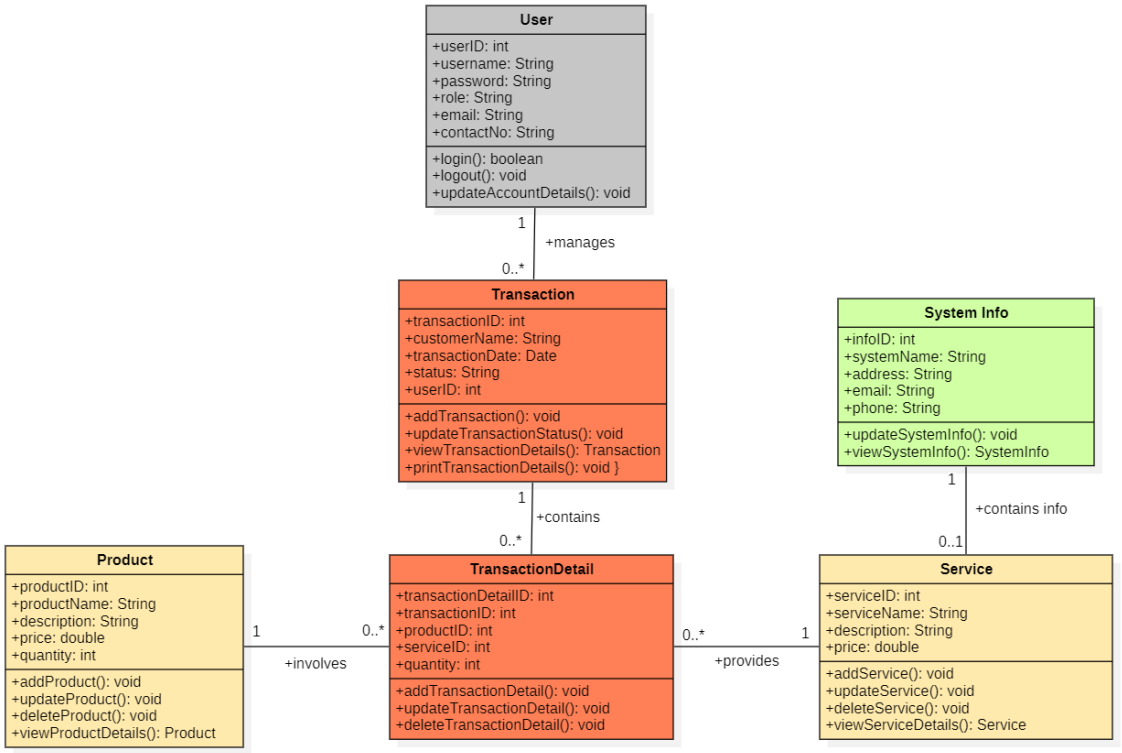
6.1 ENTITY RELATIONSHIP(ER) DIAGRAM:

The **Entity-Relationship Diagram (ERD)** for the Computer Repair Shop Management System models the relationships between different entities in the system, such as Users, Products, Services, and Transactions. It defines how these entities relate to each other through relationships like "handles," "manages," and "requests." The ERD helps in designing the database structure by clearly identifying the entities, attributes, and their connections, making it easier to manage and retrieve data efficiently. This diagram, although typically not classified as **UML**, was created in **StarUML** software for a detailed database schema design.



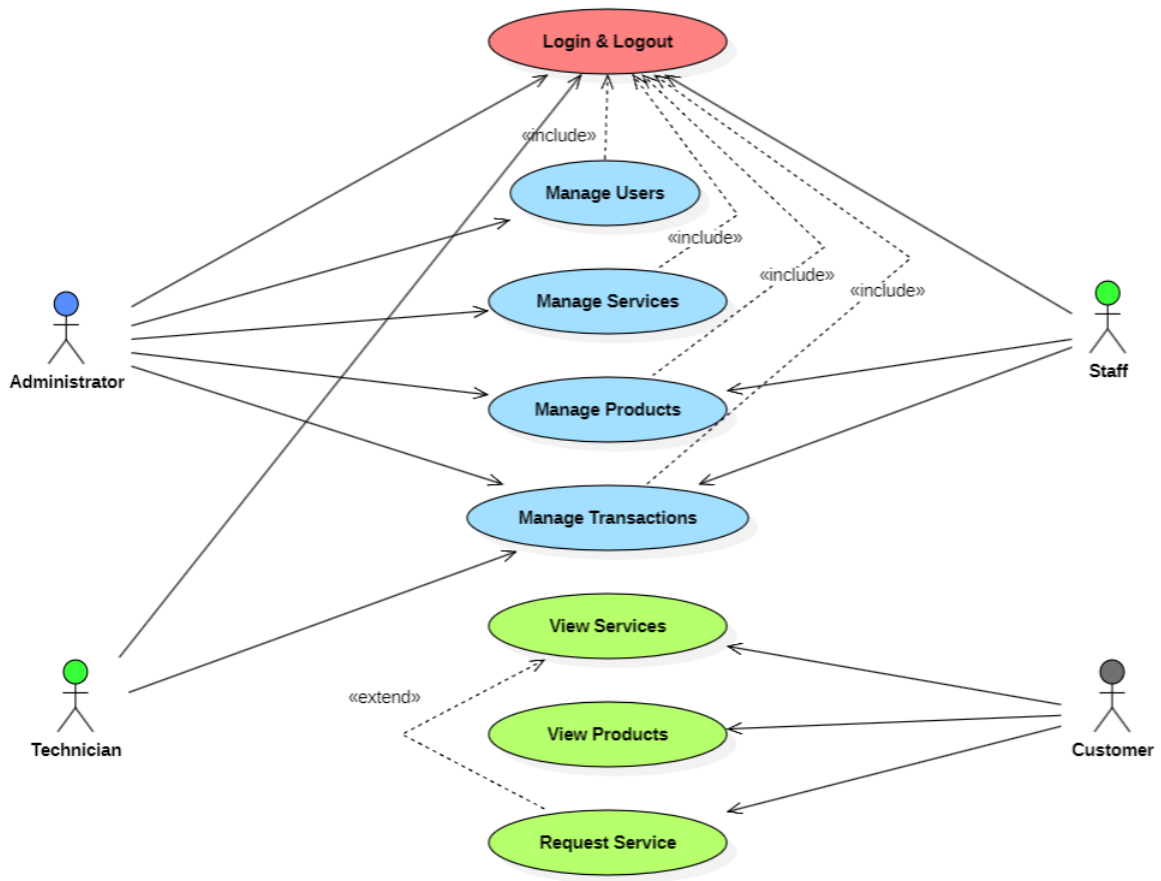
6.2 CLASS DIAGRAM :

The **Class Diagram** represents the static structure of the Computer Repair Shop Management System. It defines the system's classes, including Administrator, Staff, Technician, Customer, Product, Service, Transaction, and their relationships. Each class has attributes and methods that reflect the roles, permissions, and functionalities within the system, such as managing users, handling transactions, and managing products. The diagram provides a clear view of object-oriented design and encapsulates the relationships between different classes. This **UML** diagram was created in **StarUML** software to depict the structural design of the system.



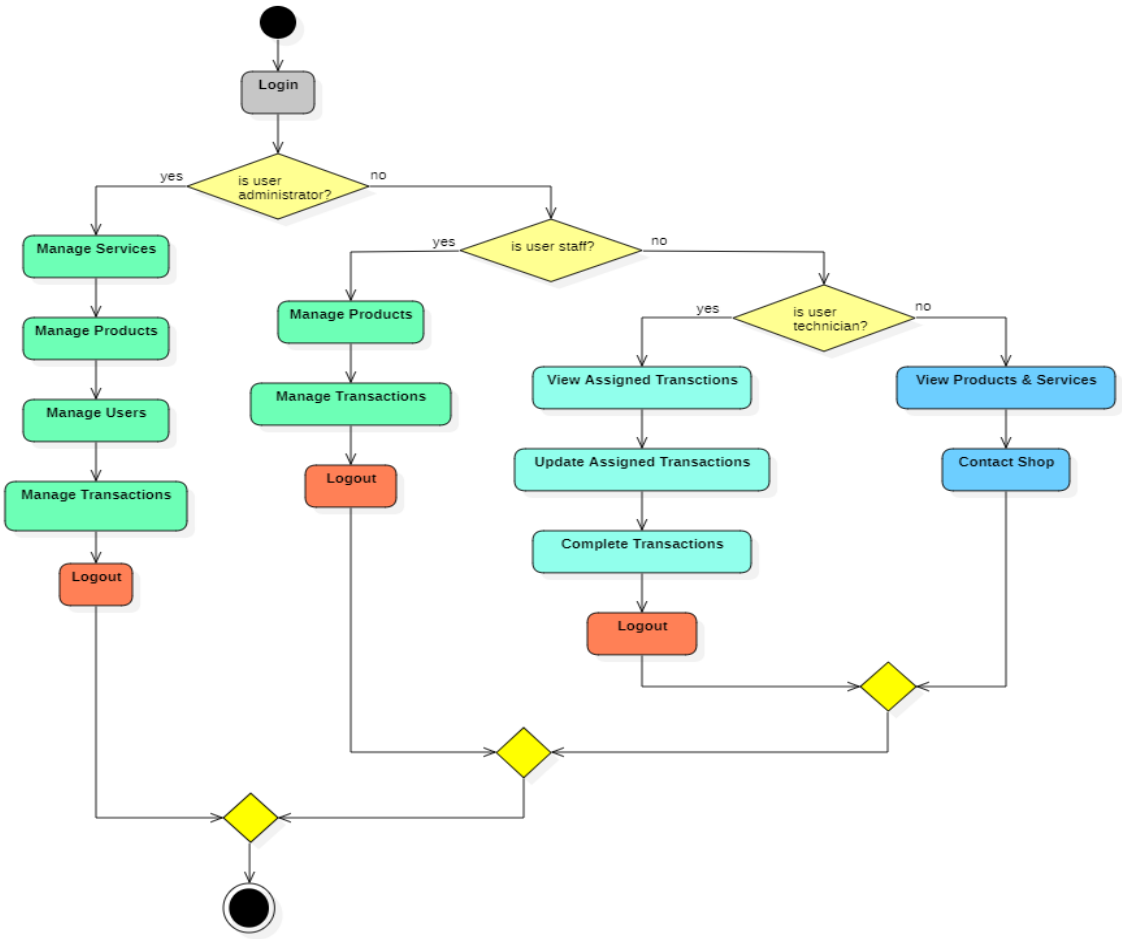
6.3 USE CASE DIAGRAM :

The **Use Case Diagram** for the Computer Repair Shop Management System provides a high-level view of the system's functionalities and the interactions between users (Administrator, Staff, Technician, and Customer) and the system. It outlines use cases like managing users, products, services, transactions, and customer interactions through the public site. The diagram shows the relationships between actors and use cases, capturing both the primary functionalities and optional actions. This **UML** diagram was developed using **StarUML** to provide a clear overview of the system's behaviour and user roles.



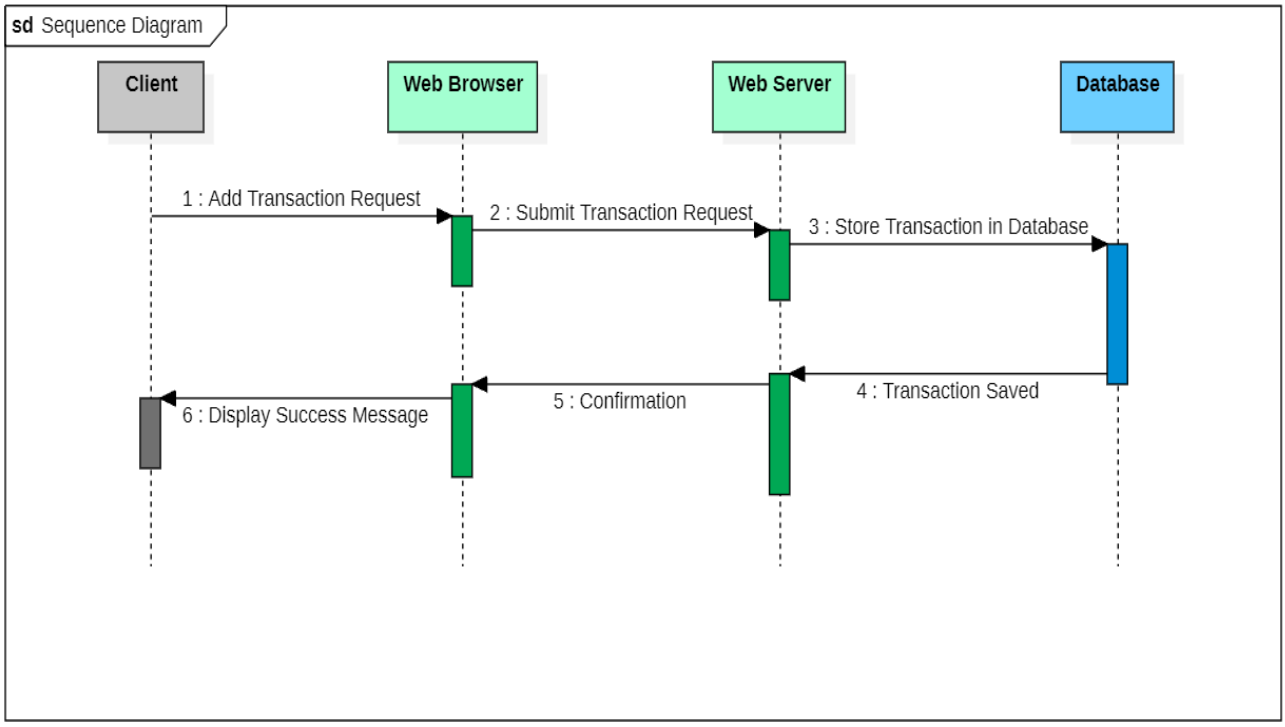
6.4 ACTIVITY DIAGRAM :

The **Activity Diagram** for the Computer Repair Shop Management System visually represents the workflow of various processes within the system. It captures the sequence of activities such as user login, managing services, products, transactions, and the overall process of requesting a repair service. Each activity is represented as a node, with decision points and parallel activities shown using branching and synchronization bars. This diagram helps in understanding the step-by-step flow of operations in the system. Being a **Unified Modeling Language (UML)** diagram, it was created in **StarUML** software to depict the dynamic aspects of the project.



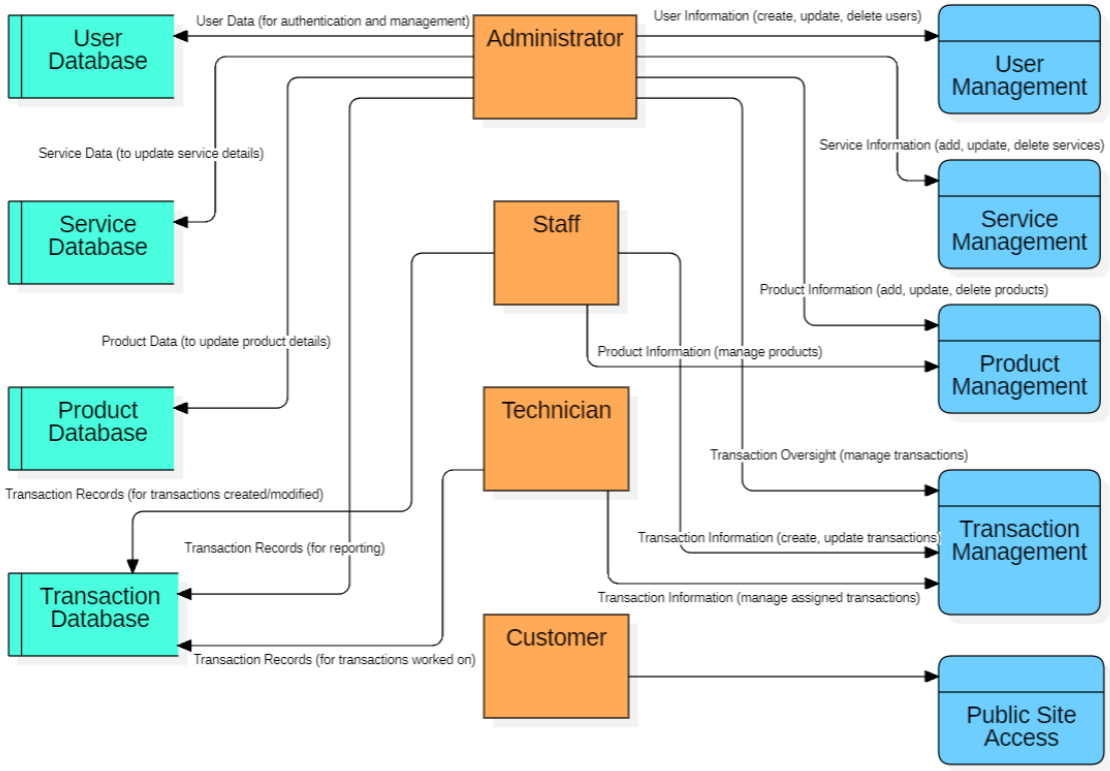
6.5 SEQUENCE DIAGRAM :

The **Sequence Diagram** for the Computer Repair Shop Management System illustrates how objects within the system interact with each other over time. It shows the sequence of messages exchanged between actors (such as Administrator, Staff, Technician, and Customer) and the system components during specific processes like login, managing transactions, or requesting a service. This **UML** diagram, created in **StarUML**, is useful for understanding the timing and order of interactions, providing clarity on how different parts of the system communicate to achieve a task.



6.6 DATA FLOW DIAGRAM:

The **Data Flow Diagram (DFD)** illustrates how data moves through the Computer Repair Shop Management System. It shows the interaction between different entities such as the Administrator, Staff, Technician, and Customer, and the flow of information between the system's processes like managing products, transactions, and services. Data stores like the MySQL database are also represented to depict how information is stored and retrieved. This diagram, although not strictly a UML diagram, was designed using **StarUML** software to model the flow of data within the system.



TEST

Testing is an essential part of the development process to ensure that the application meets all requirements and functions correctly. This section outlines the testing methodologies used.

1. Testing Environment

The testing environment was set up on a local server using XAMPP, with various web browsers (Chrome, Firefox) for compatibility testing. The system was tested on different devices, including desktop computers and mobile devices, to ensure responsiveness and usability.

2. Types of Testing

- **Unit Testing:**
  - Each function and component was tested in isolation to ensure they performed as expected. For instance, validation functions for user input were tested with various inputs to verify their correctness.
- **Integration Testing:**
  - Different modules of the application were tested together to ensure they interacted correctly. For example, the service management module was tested in conjunction with the transaction management module to ensure that services could be correctly added to transactions.
- **System Testing:**
  - The entire system was tested as a whole. This included end-to-end testing of user workflows, such as logging in, managing services, and completing transactions.
- **Functional Testing:**
  - Functional tests were conducted to ensure all key features worked as expected. This included testing login/logout functionality, adding/removing services, and generating printable transaction receipts.

- **Usability Testing:**
  - User interface testing was performed to assess how easily users could navigate the system. Feedback was gathered from potential users to identify areas for improvement in the design and usability of the interface.
- **Security Testing:**
  - The security of the application was tested to ensure that user data was protected. This included testing for SQL injection vulnerabilities, session fixation, and ensuring proper validation and sanitization of user inputs.

TEST CASES

1. Administrator Role Testing

1.1. Home Page Testing

- **Objective**: Verify that the homepage displays the correct summary and images.
- **Preconditions**: Admin is logged in.
- **Test Steps**:
  1. Login as an Administrator.
  2. Navigate to the Home Page.
  3. Check if the summary of transactions, services, products, and users is displayed.
  4. Verify that images are correctly loaded.
- **Expected Result**: The summary should display the correct data, and all images should be visible without any errors.

1.2. Service Management

- **Objective**: Test the full functionality of service management (Add, List, View, Delete).
- **Preconditions**: Admin is logged in.

- **Test Steps**:
    1. Navigate to the "Add New Service" form.
    2. Fill in the service details (name, description, price, etc.) and submit.
    3. Navigate to the "List All Services" section and verify the new service is listed.
    4. Click on the service to view its details.
    5. Try deleting the service and confirm deletion.
- **Expected Result**:
    - New services should be added successfully.
    - The service should appear in the list and be viewable with accurate details.
    - Service should be deletable with confirmation.

1.3. Product Management

- **Objective**: Ensure product management works correctly (Add, List, View, Update, Delete).
- **Preconditions**: Admin is logged in.
- **Test Steps**:
    1. Navigate to the "Add New Product" page.
    2. Enter product details and submit the form.
    3. Verify the product is listed under "List All Products".
    4. Select the product to view details and ensure all information is accurate.
    5. Attempt to update the product's details and confirm the changes.
    6. Delete the product and confirm.
- **Expected Result**:
    - Products should be added, viewed, updated, and deleted successfully.

1.4. Transaction Management

- **Objective**: Test transaction functionality (Add, List, View, Update, Print, Delete).

- **Preconditions**: Admin is logged in.
- **Test Steps**:
    1. Create a new transaction by filling in required details.
    2. Verify that the transaction appears in the "List All Transactions".
    3. View the transaction and check that all details are correct.
    4. Update transaction details and status.
    5. Use the print functionality to generate a printable version of the transaction.
    6. Delete the transaction.
- **Expected Result**:
    - Transactions should be added, updated, viewed, printed, and deleted without errors.

1.5. User Management

- **Objective**: Verify that user management works (Add, List, View, Update, Delete).
- **Preconditions**: Admin is logged in.
- **Test Steps**:
    1. Add a new user with different roles (Staff, Technician).
    2. View the user in the "List All Users" section.
    3. Check that user details are accurate and editable.
    4. Update user credentials or details.
    5. Delete the user.
- **Expected Result**:
    - Users should be added, viewed, updated, and deleted without issues.

1.6. System Information Update

- **Objective**: Ensure system information can be updated.
- **Preconditions**: Admin is logged in.

- **Test Steps**:
    1. Navigate to the "Update System Information" section.
    2. Change system details such as the name, description, contact details, etc.
    3. Verify that the updated information reflects across the system.
- **Expected Result**:
    - System information should be updated and visible on the relevant pages.

2. Staff Role Testing

2.1. Home Page Testing

- **Objective**: Verify the summary and images for staff users.
- **Preconditions**: Staff is logged in.
- **Test Steps**:
    - Same as Admin Home Page Test (Section 1.1).
- **Expected Result**:
    - The homepage should display a summary relevant to staff users.

2.2. Product Management

- **Objective**: Ensure staff can manage products.
- **Preconditions**: Staff is logged in.
- **Test Steps**:
    - Same as Admin Product Management Test (Section 1.3).
- **Expected Result**:
    - Staff should be able to add, view, update, and delete products.

2.3. Transaction Management

- **Objective**: Verify that staff can manage transactions.
- **Preconditions**: Staff is logged in.

- **Test Steps**:
  - Same as Admin Transaction Management Test (Section 1.4).
- **Expected Result**:
  - Staff should be able to add, view, update, and delete transactions, and use the print function.

2.4. Update Account Details

- **Objective**: Ensure staff can update their account credentials.
- **Preconditions**: Staff is logged in.
- **Test Steps**:
  1. Navigate to the "Update Account Details" section.
  2. Change password and other account details.
  3. Verify changes by logging out and logging back in.
- **Expected Result**:
  - Account details should be updated successfully.

3. Technician Role Testing

3.1. Home Page Testing

- **Objective**: Verify the summary and images for technician users.
- **Preconditions**: Technician is logged in.
- **Test Steps**:
  - Same as Admin Home Page Test (Section 1.1).
- **Expected Result**:
  - The homepage should display a summary relevant to the technician.

3.2. Transaction Management

- **Objective**: Ensure technician can manage assigned transactions.
- **Preconditions**: Technician is logged in.
- **Test Steps**:

  - Create a new transaction by filling in required details.
  - Verify that the transaction appears in the "List All Transactions".
  - View the transaction and check that all details are correct.
  - Update transaction details and status.
  - Use the print functionality to generate a printable version of the transaction.
  - Delete the transaction.
- **Expected Result**:
  - Technician should be able to view, update, and print transactions.

3.3. Update Account Details

- **Objective**: Ensure technician can update their account credentials.
- **Preconditions**: Technician is logged in.
- **Test Steps**:
  - Navigate to the "Update Account Details" section.
  - Change password and other account details.
  - Verify changes by logging out and logging back in.
- **Expected Result**:
  - Technician should be able to update their account details.

4. Public-Side Testing

4.1. Home Page

- **Objective**: Ensure the homepage is displayed correctly to public users.
- **Preconditions**: None.
- **Test Steps**:
  1. Visit the website without logging in.
  2. Verify the welcome content is visible.
- **Expected Result**:

- The homepage should display the correct welcome message.

4.2. About Us Page

- **Objective**: Ensure the 'About Us' section is functional.
- **Preconditions**: None.
- **Test Steps**:
  1. Visit the About Us section.
  2. Verify that the content is displayed properly.
- **Expected Result**:
  - The About Us content should load without any errors.

4.3. Service Listing

- **Objective**: Verify that services are listed correctly.
- **Preconditions**: None.
- **Test Steps**:
  1. Visit the Services page.
  2. Verify that all services are listed with correct details.
  3. Perform a search for a specific service.
- **Expected Result**:
  - Services should be listed, and the search function should work as expected.

4.4. Product Listing

- **Objective**: Verify that products are listed correctly.
- **Preconditions**: None.
- **Test Steps**:
  1. Visit the Products page.
  2. Verify that all products are listed with correct prices and details.
  3. Perform a search for a specific product.

- **Expected Result**:
    - Products should be listed correctly, and the search function should work.

4.5. Login and Logout Testing

- **Objective**: Ensure that public users cannot log in.
- **Preconditions**: None.
- **Test Steps**:
    1. Attempt to log in from the public site.
- **Expected Result**:
    - Login should not be accessible to public users.

The testing phase of the **Computer Repair Shop Management System** is critical in ensuring the application meets the specified requirements and functions smoothly across all user roles. Through a combination of functional and non-functional testing, the development team meticulously verifies each feature, identifying and addressing any issues that arise. User acceptance testing (UAT) further enriches this process, allowing real users to interact with the system and provide valuable feedback, which helps refine the application before deployment.

Ultimately, this comprehensive testing approach not only enhances the reliability and efficiency of the management system but also ensures a positive user experience. By prioritizing thorough testing, the team lays a strong foundation for the application's success, enabling the computer repair shop to streamline operations and improve customer service. With a well-tested system, users can confidently manage transactions, services, and products, ensuring the application delivers its intended benefits effectively.

CODE

FRONTEND:

1. Welcome.html (Home page)

A welcome page for users visiting the website.

Code:

```
<p style="margin-right: 0px; margin-bottom: 0px; margin-left: 0px; text-align: center; padding: 0px; font-family: DauphinPlain; font-size: 70px; line-height: 90px;"></p><h2 style="text-align: center; margin: 15px; padding: 0px; position: relative;"><span style="font-family: Impact;">Welcome to TechCare Computer Repair &amp;amp; Parts</span></h2><p style="margin-right: 0px; margin-bottom: 15px; margin-left: 0px; padding: 0px;"><span style="font-family: Helvetica;"><b>At TechCare Computer Repair &amp;amp; Parts, we specialize in offering high-quality repair services for all types of computers, whether it's a laptop or desktop, PC or Mac. Our experienced technicians are equipped to handle a wide range of issues, from software glitches and malware removal to hardware repairs like screen replacements and motherboard diagnostics. We pride ourselves on delivering fast, reliable services to ensure minimal downtime for our customers. Whether you're dealing with a slow system, a broken component, or need data recovery, we are here to get your devices running smoothly again.</b></span></p><p style="margin-right: 0px; margin-bottom: 15px; margin-left: 0px; padding: 0px;"><span style="font-family: Helvetica;"><b>In addition to repair services, we also stock a comprehensive range of computer hardware parts. From graphics cards and processors to memory modules and power supplies, we carry trusted brands to suit both casual users and tech enthusiasts. Whether you are upgrading your current setup, building a new system from scratch, or simply need a replacement part, our knowledgeable staff is ready to help you choose the right components to meet your needs. At TechCare, we strive to be your one-stop solution for all things computer-related, ensuring both quality repairs and access to top-tier hardware.</b></span></p>
```

2. about.html (About page)

Displays the "About Us" section of the public-facing site.

Code:

** About Us <hr style="margin: 0px; padding: 0px; clear: both; border-top: 0px; height: 1px; background-image: linear-gradient(to right, rgba(0, 0, 0, 0), rgba(0, 0, 0, 0.75), rgba(0, 0, 0, 0));"> <div id="bannerL" style="margin: 0px 0px 0px -160px; padding: 0px; position: sticky; top: 20px; width: 160px; height: 10px; float: left; text-align: right; color: rgb(0, 0, 0); font-family: "Open Sans", Arial, sans-serif; font-size: 14px; background-color: rgb(255, 255, 255);"> <div id="bannerR" style="margin: 0px -160px 0px 0px; padding: 0px; position: sticky; top: 20px; width: 160px; height: 10px; float: right; color: rgb(0, 0, 0); font-family: "Open Sans", Arial, sans-serif; font-size: 14px; background-color: rgb(255, 255, 255);"> <div class="boxed" style="margin: 10px 28.7969px; padding: 0px; clear: both; color: rgb(0, 0, 0); font-family: "Open Sans", Arial, sans-serif; font-size: 14px; text-align: center; background-color: rgb(255, 255, 255);"> Welcome to TechCare Computer Repair & Parts, your trusted partner for all things tech! Established in 2009, we are a dedicated team of computer enthusiasts and skilled technicians committed to providing top-notch repair services and hardware solutions. Our mission is to keep your devices running smoothly while helping you find the best equipment to meet your needs. Whether you're a casual user or a tech professional, we have the expertise and resources to assist you. At TechCare Computer Repair & Parts, we understand that technology can sometimes be overwhelming. That's why we pride ourselves on offering personalized service tailored to each customer's unique situation. From diagnosing hardware issues to optimizing system performance, our technicians utilize the latest tools and techniques to ensure your devices are in excellent condition. We believe in transparency, providing clear explanations of the repair process and costs, so you can make**

**informed decisions. In addition to our repair services, we also offer a wide selection of high-quality hardware and accessories. Whether you're looking to upgrade your gaming rig, enhance your office setup, or simply find a reliable laptop, our knowledgeable staff is here to guide you through every step of the process. We carefully curate our inventory to feature only the best brands and products, ensuring you get the performance and reliability you deserve. We believe in building lasting relationships with our customers, which is why we prioritize excellent customer service and ongoing support. Our team is always ready to answer your questions, offer advice, and provide assistance even after your purchase or repair is complete. We want you to feel confident and empowered in your technology choices, and our commitment to your satisfaction is what sets us apart. When you choose TechCare Computer Repair & Parts, you're not just getting a service—you're joining a community of tech lovers who share your passion for innovation and quality. Let us be your go-to destination for all your computer repair and hardware needs! Community is at the heart of what we do. We actively support local initiatives and aim to foster a tech-savvy environment by hosting workshops and training sessions. At TechCare Computer Repair & Parts, we're not just a repair shop; we're a place where technology meets community. Thank you for choosing us to serve your tech needs—we look forward to helping you stay connected! **

3. index.php (Main entry point)

This file is typically the main entry point of the website. Including it will provide a clear idea of how the system initializes and directs the user based on their role (public, admin, staff, or technician).

Code:

```
<?php require_once('config.php'); ?>
<!DOCTYPE html>
<html lang="en">
<?php require_once('inc/header.php') ?>
<?php if($_settings->chk_flashdata('success')): ?>
<script>
 $(function(){
   alert_toast("<?php echo $_settings->flashdata('success') ?>",'success')
 })
</script>
<?php endif;?>
<body>
<?php require_once('inc/topBarNav.php') ?>
<?php $page = isset($_GET['p']) ? $_GET['p'] : 'home';  ?>
<?php
   if(!file_exists($page.".php") && !is_dir($page)){
      include '404.html';
   }else{
   if(is_dir($page))
      include $page.'/index.php';
   else
      include $page.'.php';

   }
?>
<?php require_once('inc/footer.php') ?>

  <div class="modal fade" id="uni_modal" role='dialog'>
    <div class="modal-dialog   rounded-0 modal-md modal-dialog-centered"
role="document">
```

```
    <div class="modal-content  rounded-0">
      <div class="modal-header">
      <h5 class="modal-title"></h5>
    </div>
    <div class="modal-body">
    </div>
    <div class="modal-footer">
      <button type="button" class="btn btn-primary" id='submit'
onclick="$('#uni_modal form').submit()">Save</button>
      <button type="button" class="btn btn-secondary" data-
dismiss="modal">Cancel</button>
    </div>
    </div>
   </div>
  </div>
  <div class="modal fade" id="uni_modal_right" role='dialog'>
   <div class="modal-dialog  rounded-0 modal-full-height  modal-md"
role="document">
    <div class="modal-content rounded-0">
      <div class="modal-header">
      <h5 class="modal-title">></h5>
      <button type="button" class="close" data-dismiss="modal" aria-
label="Close">
        <span class="fa fa-arrow-right">></span>
      </button>
    </div>
    <div class="modal-body">
    </div>
    </div>
   </div>
  </div>
  <div class="modal fade" id="viewer_modal" role='dialog'>
   <div class="modal-dialog modal-md" role="document">
    <div class="modal-content">
          <button type="button" class="btn-close" data-dismiss="modal"><span
class="fa fa-times"></span></button>
          <img src="" alt="">
```

```
      </div>
    </div>
  </div>
  <div class="modal fade" id="confirm_modal" role='dialog'>
    <div class="modal-dialog modal-md modal-dialog-centered"
role="document">
      <div class="modal-content">
        <div class="modal-header">
        <h5 class="modal-title">Confirmation</h5>
      </div>
      <div class="modal-body">
        <div id="delete_content"></div>
      </div>
      <div class="modal-footer">
        <button type="button" class="btn btn-primary" id='confirm'
onclick="">Continue</button>
        <button type="button" class="btn btn-secondary" data-
dismiss="modal">Close</button>
      </div>
      </div>
    </div>
  </div>
</body>
</html>
```

4. navigation.php (Navigation structure)

Handles the navigation structure of the admin panel.

Code:

```
</style>
<!-- Main Sidebar Container -->
    <aside class="main-sidebar sidebar-dark-primary elevation-4 sidebar-no-expand">
      <!-- Brand Logo -->
      <a href="<?php echo base_url ?>admin" class="brand-link bg-gradient-navy text-sm">
      <img src="<?php echo validate_image($_settings->info('logo'))?>" alt="Store Logo" class="brand-image img-circle elevation-3" style="opacity: .8;width: 1.5rem;height: 1.5rem;max-height: unset">
      <span class="brand-text font-weight-light"><?php echo $_settings->info('short_name') ?></span>
      </a>
      <!-- Sidebar -->
      <div class="sidebar os-host os-theme-light os-host-overflow os-host-overflow-y os-host-resize-disabled os-host-transition os-host-scrollbar-horizontal-hidden">
        <div class="os-resize-observer-host observed">
          <div class="os-resize-observer" style="left: 0px; right: auto;"></div>
        </div>
        <div class="os-size-auto-observer observed" style="height: calc(100% + 1px); float: left;">
          <div class="os-resize-observer"></div>
        </div>
        <div class="os-content-glue" style="margin: 0px -8px; width: 249px; height: 646px;"></div>
        <div class="os-padding">
          <div class="os-viewport os-viewport-native-scrollbars-invisible" style="overflow-y: scroll;">
            <div class="os-content" style="padding: 0px 8px; height: 100%; width: 100%;">
              <!-- Sidebar user panel (optional) -->
              <div class="clearfix"></div>
              <!-- Sidebar Menu -->
              <nav class="mt-4">
```

```
            <ul class="nav nav-pills nav-sidebar flex-column text-sm nav-
compact nav-flat nav-child-indent nav-collapse-hide-child" data-
widget="treeview" role="menu" data-accordion="false">
            <li class="nav-item dropdown">
              <a href="./" class="nav-link nav-home">
                <i class="nav-icon fas fa-tachometer-alt"></i>
                <p>
                  Dashboard
                </p>
              </a>
            </li>
            <?php if($_settings->userdata('type') != 3): ?>
            <li class="nav-item dropdown">
              <a href="<?php echo base_url ?>admin/?page=products"
class="nav-link nav-products">
                <i class="nav-icon fas fa-microchip"></i>
                <p>
                  Product List
                </p>
              </a>
            </li>
            <?php endif; ?>
            <li class="nav-item dropdown">
              <a href="<?php echo base_url ?>admin/?page=transactions"
class="nav-link nav-transactions">
                <i class="nav-icon fas fa-clipboard-list"></i>
                <p>
                  Transactions
                </p>
              </a>
            </li>
            <?php if($_settings->userdata('type') == 1): ?>
            <li class="nav-header">Maintenance</li>
            <li class="nav-item dropdown">
              <a href="<?php echo base_url ?>admin/?page=services"
class="nav-link nav-services">
                <i class="nav-icon fas fa-cogs"></i>
```

```
                <p>
                  Service List
                </p>
              </a>
            </li>
            <li class="nav-item dropdown">
              <a href="<?php echo base_url ?>admin/?page=user/list"
class="nav-link nav-user/list">
                <i class="nav-icon fas fa-users-cog"></i>
                <p>
                  User List
                </p>
              </a>
            </li>
            <li class="nav-item dropdown">
              <a href="<?php echo base_url ?>admin/?page=system_info"
class="nav-link nav-system_info">
                <i class="nav-icon fas fa-tools"></i>
                <p>
                  Settings
                </p>
              </a>
            </li>
            <?php endif; ?>
          </ul>
        </nav>
        <!-- /.sidebar-menu -->
      </div>
    </div>
  </div>
  <div class="os-scrollbar os-scrollbar-horizontal os-scrollbar-unusable os-
scrollbar-auto-hidden">
    <div class="os-scrollbar-track">
      <div class="os-scrollbar-handle" style="width: 100%; transform:
translate(0px, 0px);"></div>
    </div>
  </div>
```

```
        <div class="os-scrollbar os-scrollbar-vertical os-scrollbar-auto-hidden">
          <div class="os-scrollbar-track">
            <div class="os-scrollbar-handle" style="height: 55.017%; transform:
translate(0px, 0px);"></div>
          </div>
        </div>
        <div class="os-scrollbar-corner"></div>
      </div>
      <!-- /.sidebar -->
    </aside>
    <script>
   $(document).ready(function(){
    var page = '<?php echo isset($_GET['page']) ? $_GET['page'] : 'home' ?>';
    var s = '<?php echo isset($_GET['s']) ? $_GET['s'] : '' ?>';
    page = page.replace(/\/g,'_');
    console.log(page)

    if($('.nav-link.nav-'+page).length > 0){
          $('.nav-link.nav-'+page).addClass('active')
      if($('.nav-link.nav-'+page).hasClass('tree-item') == true){
          $('.nav-link.nav-'+page).closest('.nav-
treeview').siblings('a').addClass('active')
        $('.nav-link.nav-'+page).closest('.nav-treeview').parent().addClass('menu-
open')
      }
      if($('.nav-link.nav-'+page).hasClass('nav-is-tree') == true){
        $('.nav-link.nav-'+page).parent().addClass('menu-open')
      }

    }
    $('.nav-link.active').addClass('bg-gradient-navy')
   })
  </script>
```

5. topBarNav.php (Top navigation mark)

Manages the top navigation bar in the admin interface.

Code:

```
<nav class="navbar navbar-expand-lg navbar-dark bg-gradient-navy">
        <div class="container px-4 px-lg-5 ">
            <button class="navbar-toggler btn btn-sm" type="button" data-toggle="collapse" data-target="#navbarSupportedContent" aria-controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation"><span class="navbar-toggler-icon"></span></button>
            <a class="navbar-brand" href="./">
            <img src="<?php echo validate_image($_settings->info('logo')) ?>" width="30" height="30" class="d-inline-block align-top" alt="" loading="lazy">
            <?php echo $_settings->info('short_name') ?>
            </a>


            <div class="collapse navbar-collapse" id="navbarSupportedContent">
              <ul class="navbar-nav me-auto mb-2 mb-lg-0 ms-lg-4">
                <li class="nav-item"><a class="nav-link text-white" aria-current="page" href="./">Home</a></li>
                <li class="nav-item"><a class="nav-link text-white" aria-current="page" href="./?p=services">Our Services</a></li>
                <li class="nav-item"><a class="nav-link text-white" aria-current="page" href="./?p=products">Our Products</a></li>
                <li class="nav-item"><a class="nav-link text-white" href="./?p=about">About</a></li>
              </ul>
              <div class="d-flex align-items-center">
                <a class="font-weight-bolder text-light mx-2 text-decoration-none" href="./admin">Login</a>
              </div>
            </div>
        </div>
      </nav>
<script>
 $(function(){
   $('#login-btn').click(function(){
     uni_modal("","login.php")
   })
```

```
  $('#navbarResponsive').on('show.bs.collapse', function () {
     $('#mainNav').addClass('navbar-shrink')
  })
  $('#navbarResponsive').on('hidden.bs.collapse', function () {
     if($('body').offset.top == 0)
       $('#mainNav').removeClass('navbar-shrink')
  })
 })

 $('#search-form').submit(function(e){
  e.preventDefault()
   var sTxt = $('[name="search"]').val()
   if(sTxt != '')
    location.href = './?p=products&search='+sTxt;
 })
</script>
```

BACKEND:

1. login.php (Login functionality)

Since login is central to the role-based system, showcasing this code will give an understanding of how user authentication works. It focuses on the validation of user credentials, session management, and redirection after successful/failed login attempts.

Code:

```
<?php require_once('../config.php') ?>
<!DOCTYPE html>
<html lang="en" class="" style="height: auto;">
 <?php require_once('inc/header.php') ?>
<body class="hold-transition login-page">
  <script>
   start_loader()
  </script>
  <style>
   body{
     background-image: url("<?php echo validate_image($_settings-
>info('cover')) ?>");
     background-size:cover;
     background-repeat:no-repeat;
     backdrop-filter: contrast(1);
   }
   #page-title{
     text-shadow: 6px 4px 7px black;
     font-size: 3.5em;
     color: #fff4f4 !important;
     background: #8080801c;
   }
  </style>
  <h1 class="text-center text-white px-4 py-5" id="page-title"><b><?php echo
$_settings->info('name') ?></b></h1>
<div class="login-box">
  <!-- /.login-logo -->
  <div class="card card-navy my-2">
```

```
  <div class="card-body">
    <p class="login-box-msg">Please enter your credentials</p>
    <form id="login-frm" action="" method="post">
      <div class="input-group mb-3">
        <input type="text" class="form-control" name="username" autofocus
placeholder="Username">
        <div class="input-group-append">
          <div class="input-group-text">
            <span class="fas fa-user"></span>
          </div>
        </div>
      </div>
      <div class="input-group mb-3">
        <input type="password" class="form-control"  name="password"
placeholder="Password">
        <div class="input-group-append">
          <div class="input-group-text">
            <span class="fas fa-lock"></span>
          </div>
        </div>
      </div>
      <div class="row">
        <div class="col-8">
          <a href="<?php echo base_url ?>">Go to Website</a>
        </div>
        <!-- /.col -->
        <div class="col-4">
          <button type="submit" class="btn btn-primary btn-block">Sign
In</button>
        </div>
        <!-- /.col -->
      </div>
    </form>
  </div>
  <!-- /.card-body -->
 </div>
```

```
  <!-- /.card -->
</div>
<!-- /.login-box -->

<!-- jQuery -->
<script src="plugins/jquery/jquery.min.js"></script>
<!-- Bootstrap 4 -->
<script src="plugins/bootstrap/js/bootstrap.bundle.min.js"></script>
<!-- AdminLTE App -->
<script src="dist/js/adminlte.min.js"></script>

<script>
  $(document).ready(function(){
    end_loader();
  })
</script>
</body>
</html>
```

2. DBConnection.php or config.php (Database connection)

Include the code that handles the database connection. This is a crucial part of any dynamic web application. The file will demonstrate how the system connects to the MySQL database and manages queries.

Code:

```
<?php
if(!defined('DB_SERVER')){
   require_once("../initialize.php");
}
class DBConnection{

   private $host = DB_SERVER;
   private $username = DB_USERNAME;
   private $password = DB_PASSWORD;
   private $database = DB_NAME;

   public $conn;

   public function __construct(){
      if (!isset($this->conn)) {
         $this->conn = new mysqli($this->host, $this->username, $this-
>password, $this->database);

         if (!$this->conn) {
            echo 'Cannot connect to database server';
            exit;
         }
      }

   }
   public function __destruct(){
      $this->conn->close();
   }
}
?>
```

3. config.php (Configuration)

In addition to database credentials, config.php may include important application settings, such as error reporting levels, session management configurations, and paths to essential directories.

Code:

```
<?php
ob_start();
ini_set('date.timezone','Asia/Manila');
date_default_timezone_set('Asia/Manila');
session_start();

require_once('initialize.php');
require_once('classes/DBConnection.php');
require_once('classes/SystemSettings.php');
$db = new DBConnection;
$conn = $db->conn;
function redirect($url=''){
   if(!empty($url))
   echo '<script>location.href="'.base_url .$url.'"</script>';
}
function validate_image($file){
   if(!empty($file)){
         // exit;
      $ex = explode("?",$file);
      $file = $ex[0];
      $ts = isset($ex[1]) ? "?".$ex[1] : '';
      if(is_file(base_app.$file)){
         return base_url.$file.$ts;
      }else{
         return base_url.'dist/img/no-image-available.png';
      }
   }else{
      return base_url.'dist/img/no-image-available.png';
   }
```

```
}
function format_num($number = '' , $decimal = ''){
   if(is_numeric($number)){
      $ex = explode(".",$number);
      $decLen = isset($ex[1]) ? strlen($ex[1]) : 0;
      if(is_numeric($decimal)){
         return number_format($number,$decimal);
      }else{
         return number_format($number,$decLen);
      }
   }else{
      return "Invalid Input";
   }
}
function isMobileDevice(){
   $aMobileUA = array(
      '/iphone/i' => 'iPhone',
      '/ipod/i' => 'iPod',
      '/ipad/i' => 'iPad',
      '/android/i' => 'Android',
      '/blackberry/i' => 'BlackBerry',
      '/webos/i' => 'Mobile'
   );

   //Return true if Mobile User Agent is detected
   foreach($aMobileUA as $sMobileKey => $sMobileOS){
      if(preg_match($sMobileKey, $_SERVER['HTTP_USER_AGENT'])){
         return true;
      }
   }
   //Otherwise return false..
   return false;
}
ob_end_flush();
?>
```

4. manage_services.php & manage_products.php (Service & Product Management)

The service & product management module (add or list services and products) will showcase how CRUD operations (Create, Read, Update, Delete) are handled. This file will also demonstrate interaction with the database and display data in the user interface.

Code (manage_services.php):

```
<?php

require_once('../../config.php');
if(isset($_GET['id']) && $_GET['id'] > 0){
   $qry = $conn->query("SELECT * from `service_list` where id = '{$_GET['id']}' ");
   if($qry->num_rows > 0){
      foreach($qry->fetch_assoc() as $k => $v){
         $$k=$v;
      }
   }
}
?>
<div class="container-fluid">
   <form action="" id="service-form">
      <input type="hidden" name ="id" value="<?php echo isset($id) ? $id : ''
?>">
      <div class="form-group">
         <label for="name" class="control-label">Name</label>
         <input type="text" name="name" id="name" class="form-control form-
control-sm rounded-0" value="<?php echo isset($name) ? $name : '';
?>"  required/>
      </div>
      <div class="form-group">
         <label for="description" class="control-label">Description</label>
         <textarea type="text" name="description" id="description" class="form-
control form-control-sm rounded-0" required><?php echo isset($description) ?
$description : ''; ?></textarea>
      </div>
      <div class="form-group">
         <label for="price" class="control-label">Price</label>
```

```
        <input type="number" name="price" id="price" class="form-control
form-control-sm rounded-0 text-right" value="<?php echo isset($price) ? $price
: ''; ?>"  required/>
      </div>
      <div class="form-group">
        <label for="status" class="control-label">Status</label>
        <select name="status" id="status" class="form-control form-control-sm
rounded-0" required>
        <option value="1" <?php echo isset($status) && $status == 1 ? 'selected'
: '' ?>>Active</option>
        <option value="0" <?php echo isset($status) && $status == 0 ? 'selected'
: '' ?>>Inactive</option>
        </select>
      </div>
    </form>
</div>
<script>
    $(document).ready(function(){
      $('#service-form').submit(function(e){
        e.preventDefault();
        var _this = $(this)
         $('.err-msg').remove();
        start_loader();
        $.ajax({
          url:_base_url_+"classes/Master.php?f=save_service",
          data: new FormData($(this)[0]),
          cache: false,
          contentType: false,
          processData: false,
          method: 'POST',
          type: 'POST',
          dataType: 'json',
          error:err=>{
            console.log(err)
            alert_toast("An error occured",'error');
            end_loader();
          },
```

```
            success:function(resp){
                if(typeof resp =='object' && resp.status == 'success'){
                    location.reload()
                }else if(resp.status == 'failed' && !!resp.msg){
                    var el = $('<div>')
                        el.addClass("alert alert-danger err-msg").text(resp.msg)
                        _this.prepend(el)
                        el.show('slow')
                        $("html, body").animate({ scrollTop:
_this.closest('.card').offset().top }, "fast");
                        end_loader()
                }else{
                    alert_toast("An error occured",'error');
                    end_loader();
                    console.log(resp)
                }
            }
        })
    })

  })
</script>
```

Code (manage_products.php):

```
<?php
require_once('../../config.php');
if(isset($_GET['id']) && $_GET['id'] > 0){
  $qry = $conn->query("SELECT * from `product_list` where id = '{$_GET['id']}'
");
  if($qry->num_rows > 0){
    foreach($qry->fetch_assoc() as $k => $v){
      $$k=$v;
    }
```

```
    }
}
?>
<style>
    #cimg{
        width:100%;
        max-height:20vh;
        object-fit:scale-down;
        object-position:center center;
    }
</style>
<div class="container-fluid">
    <form action="" id="product-form">
        <input type="hidden" name ="id" value="<?php echo isset($id) ? $id : ''
?>">
        <div class="form-group">
            <label for="name" class="control-label">Name</label>
            <input type="text" name="name" id="name" class="form-control form-
control-sm rounded-0" value="<?php echo isset($name) ? $name : '';
?>"  required/>
        </div>
        <div class="form-group">
            <label for="description" class="control-label">Description</label>
            <textarea type="text" name="description" id="description" class="form-
control form-control-sm rounded-0" required><?php echo isset($description) ?
$description : ''; ?></textarea>
        </div>
        <div class="form-group">
            <label for="price" class="control-label">Price</label>
            <input type="number" name="price" id="price" class="form-control
form-control-sm rounded-0 text-right" value="<?php echo isset($price) ? $price
: ''; ?>"  required/>
        </div>
        <div class="form-group">
            <label for="status" class="control-label">Status</label>
            <select name="status" id="status" class="form-control form-control-sm
rounded-0" required>
                <option value="1" <?php echo isset($status) && $status == 1 ?
'selected' : '' ?>>Active</option>
```

```
            <option value="0" <?php echo isset($status) && $status == 0 ?
'selected' : '' ?>>Inactive</option>
        </select>
    </div>
    <div class="form-group">
        <label for="" class="control-label">Thumbnail</label>
        <div class="custom-file">
            <input type="file" class="custom-file-input rounded-circle"
id="customFile1" accept="image/png, image/jpeg" name="img"
onchange="displayImg(this,$(this))">
            <label class="custom-file-label" for="customFile1">Choose
file</label>
        </div>
    </div>
    <div class="form-group">
        <img src="<?= validate_image(isset($image_path) ? $image_path : '') ?>"
alt="" id="cimg" class="img-fluid img-thumbnail w-100">
    </div>
  </form>
</div>
<script>
  function displayImg(input,_this) {
    if (input.files && input.files[0]) {
        var reader = new FileReader();
        reader.onload = function (e) {
            $('#cimg').attr('src', e.target.result);
            _this.siblings('.custom-file-label').html(input.files[0].name)
        }
        reader.readAsDataURL(input.files[0]);
    }else{
            $('#cimg').attr('src', "<?= validate_image(isset($image_path) ?
$image_path : '') ?>");
    }
  }
  $(document).ready(function(){
    $('#product-form').submit(function(e){
        e.preventDefault();
        var _this = $(this)
```

```
            $('.err-msg').remove();
            start_loader();
            $.ajax({
                url:_base_url_+"classes/Master.php?f=save_product",
                data: new FormData($(this)[0]),
                cache: false,
                contentType: false,
                processData: false,
                method: 'POST',
                type: 'POST',
                dataType: 'json',
                error:err=>{
                    console.log(err)
                    alert_toast("An error occured",'error');
                    end_loader();
                },
                success:function(resp){
                    if(typeof resp ='object' && resp.status == 'success'){
                        location.reload()
                    }else if(resp.status == 'failed' && !!resp.msg){
                        var el = $('<div>')
                            el.addClass("alert alert-danger err-msg").text(resp.msg)
                            _this.prepend(el)
                            el.show('slow')
                            $("html, body,.modal").scrollTop(0);
                            end_loader()
                    }else{
                        alert_toast("An error occured",'error');
                        end_loader();
                    }
                }
            })
        })
    })
</script>
```

5. manage_transaction.php (Transaction Management)

Transactions are a key feature of the system, and including one of these files will highlight how transactions are displayed, updated, and deleted. This is relevant for showing the core business logic related to computer repair management.

Code:

```
<?php
if(isset($_GET['id'])){
   $qry = $conn->query("SELECT * FROM `transaction_list` where id =
'{$_GET['id']}' ");
   if($qry->num_rows > 0){
      $res = $qry->fetch_array();
      foreach($res as $k => $v){
         if(!is_numeric($k)){
            $$k = $v;
         }
      }
   }else{
      echo '<script> alert("Unknown Transaction\'s ID.");
location.replace("./?page=transactions"); </script>';
   }
}
?>
<div class="content py-3">
   <div class="container-fluid">
      <div class="card card-outline card-outline rounded-0 shadow blur">
         <div class="card-header">
            <h5 class="card-title"><?= isset($id) ? "Update ". $code . "
Transaction" : "New Transaction" ?></h5>
         </div>
         <div class="card-body">
            <div class="container-fluid">
               <form action="" id="transaction-form">
                  <input type="hidden" name="id" value="<?= isset($id) ? $id : ''
?>">
                  <input type="hidden" name="amount" value="<?=
isset($amount) ? $amount : '' ?>">
                  <div class="row">
```

```
                <div class="col-lg-12 col-md-12 col-sm-12 col-xs-12">
                  <div class="form-group mb-3">
                    <label for="client_name" class="control-label">Client Full
Name</label>
                    <input type="text" name="client_name" id="client_name"
class="form-control form-control-sm rounded-0" value="<?=
isset($client_name) ? $client_name : "" ?>" required="required">
                  </div>
                </div>
              </div>
              <div class="row">
                <div class="col-lg-6 col-md-6 col-sm-12 col-xs-12">
                  <div class="form-group mb-3">
                    <label for="contact" class="control-label">Client Contact
#</label>
                    <input type="text" name="contact" id="contact"
class="form-control form-control-sm rounded-0" value="<?= isset($contact) ?
$contact : "" ?>" required="required">
                  </div>
                </div>
                <div class="col-lg-6 col-md-6 col-sm-12 col-xs-12">
                  <div class="form-group mb-3">
                    <label for="email" class="control-label">Client
Email</label>
                    <input type="email" name="email" id="email"
class="form-control form-control-sm rounded-0" value="<?= isset($email) ?
$email : "" ?>" required="required">
                  </div>
                </div>
              </div>
              <div class="row">
                <div class="col-lg-12 col-md-12 col-sm-12 col-xs-12">
                  <div class="form-group mb-3">
                    <label for="address" class="control-
label">Address</label>
                    <textarea name="address" id="address" class="form-
control form-control-sm rounded-0" required="required"><?= isset($address) ?
$address : "" ?></textarea>
                  </div>
                </div>
```

```
                </div>
                <hr>
                <div class="row">
                  <div class="col-lg-5 col-md-5 col-sm-12 col-xs-12">
                    <fieldset>
                      <legend>Services</legend>
                      <div class="row align-items-end">
                        <div class="col-lg-10 col-md-10 col-sm-12 col-xs-12">
                          <div class="form-group mb-0">
                            <label for="service_sel" class="control-
label">Select Service</label>
                            <select id="service_sel" class="form-control form-
control-sm rounded">
                              <option value="" disabled selected></option>
                              <?php
                              $service_qry = $conn->query("SELECT * FROM
`service_list` where delete_flag = 0 and `status` = 1 order by `name`");
                              while($row = $service_qry->fetch_assoc()):
                              ?>
                              <option value="<?= $row['id'] ?>" data-price =
"<?= $row['price'] ?>"><?= $row['name'] ?></option>
                              <?php endwhile; ?>
                            </select>
                          </div>
                        </div>
                        <div class="col-lg-2 col-md-2 col-sm-12 col-xs-12">
                          <button class="btn btn-default bg-gradient-navy btn-
sm rounded-0" type="button" id="add_service"><i class="fa fa-plus"></i>
Add</button>
                        </div>
                      </div>
                      <div class="clear-fix mb-2"></div>
                      <table class="table table-striped table-bordered"
id="service-list">
                        <colgroup>
                          <col width="10%">
                          <col width="60%">
                          <col width="30%">
```

```
                    </colgroup>
                    <thead>
                        <tr class="bg-gradient-navy">
                            <th class="text-center"></th>
                            <th class="text-center">Service</th>
                            <th class="text-center">Price</th>
                        </tr>
                    </thead>
                    <tbody>
                        <?php
                        $service_amount = 0;
                        if(isset($id)):
                        $ts_qry = $conn->query("SELECT ts.*, s.name as
`service` FROM `transaction_services` ts inner join `service_list` s on ts.service_id
= s.id where ts.`transaction_id` = '{$id}' ");
                        while($row = $ts_qry->fetch_assoc()):
                            $service_amount += $row['price'];
                        ?>
                        <tr>
                            <td class="text-center">
                                <button class="btn btn-outline-danger btn-sm
rounded-0 rem-service" type="button"><i class="fa fa-times"></i></button>
                            </td>
                            <td>
                                <input type="hidden" name="service_id[]"
value="<?= $row['service_id'] ?>">
                                <input type="hidden" name="service_price[]"
value="<?= $row['price'] ?>">
                                <span class="service_name"><?= $row['service']
?></span>
                            </td>
                            <td class="text-right service_price"><?=
format_num($row['price']) ?></td>
                        </tr>
                        <?php endwhile; ?>
                        <?php endif; ?>
                    </tbody>
                    <tfoot>
```

```
                    <tr class="bg-gradient-secondary">
                        <th colspan="2" class="text-center">Total</th>
                        <th class="text-right" id="service_total"><?= isset($service_amount) ? format_num($service_amount): 0 ?></th>
                    </tr>
                </tfoot>
            </table>
        </fieldset>
    </div>
    <div class="col-lg-7 col-md-7 col-sm-12 col-xs-12">
        <fieldset>
            <legend>Products</legend>
            <div class="row align-items-end">
                <div class="col-lg-10 col-md-10 col-sm-12 col-xs-12">
                    <div class="form-group mb-0">
                        <label for="product_sel" class="control-label">Select Product</label>
                        <select id="product_sel" class="form-control form-control-sm rounded">
                            <option value="" disabled selected></option>
                            <?php
                            $product_qry = $conn->query("SELECT * FROM `product_list` where delete_flag = 0 and `status` = 1 order by `name`");
                            while($row = $product_qry->fetch_assoc()):
                            ?>
                            <option value="<?= $row['id'] ?>" data-price = "<?= $row['price'] ?>"><?= $row['name'] ?></option>
                            <?php endwhile; ?>
                        </select>
                    </div>
                </div>
                <div class="col-lg-2 col-md-2 col-sm-12 col-xs-12">
                    <button class="btn btn-default bg-gradient-navy btn-sm rounded-0" type="button" id="add_product"><i class="fa fa-plus"></i> Add</button>
                </div>
            </div>
            <div class="clear-fix mb-2"></div>
```

```
                    <table class="table table-striped table-bordered"
id="product-list">
                        <colgroup>
                            <col width="5%">
                            <col width="40%">
                            <col width="15%">
                            <col width="20%">
                            <col width="20%">
                        </colgroup>
                        <thead>
                            <tr class="bg-gradient-navy">
                                <th class="text-center"></th>
                                <th class="text-center">Item Name</th>
                                <th class="text-center">Qty</th>
                                <th class="text-center">Price</th>
                                <th class="text-center">Total</th>
                            </tr>
                        </thead>
                        <tbody>
                        <?php
                            $product_total = 0;
                            if(isset($id)):
                            $tp_qry = $conn->query("SELECT tp.*, p.name as
`product` FROM `transaction_products` tp inner join `product_list` p on
tp.product_id = p.id where tp.`transaction_id` = '{$id}' ");
                            while($row = $tp_qry->fetch_assoc()):
                                $product_total += ($row['price'] * $row['qty']);
                        ?>
                            <tr>
                                <td class="text-center">
                                    <button class="btn btn-outline-danger btn-sm
rounded-0 rem-product" type="button"><i class="fa fa-times"></i></button>
                                </td>
                                <td>
                                    <input type="hidden" name="product_id[]"
value="<?= $row['product_id'] ?>">
                                    <input type="hidden" name="product_price[]"
value="<?= $row['price'] ?>">
```

```
                                <span class="product_name"><?=
$row['product'] ?></span>
                            </td>
                            <td class=""><input type="number" min="1"
class="form-control form-control-sm rounded-0 text-right"
name="product_qty[]" value="<?= $row['qty'] ?>"></td>
                            <td class="text-right product_price"><?=
$row['price'] ?></td>
                            <td class="text-right product_total"><?=
format_num($row['price'] * $row['qty']) ?></td>
                        </tr>
                    <?php endwhile; ?>
                    <?php endif; ?>
                    </tbody>
                    <tfoot>
                        <tr class="bg-gradient-secondary">
                            <th colspan="4" class="text-center">Total</th>
                            <th class="text-right" id="product_total"><?=
isset($product_total) ? format_num($product_total): 0 ?></th>
                        </tr>
                    </tfoot>
                </table>
            </fieldset>
        </div>
    </div>
    <div class="clear-fix mb-3"></div>
    <h2 class="text-navy text-right">Total Payable Amount: <b
id="amount"><?= isset($amount) ? format_num($amount) : "0.00"
?></b></h2>
    <hr>
    <?php if($_settings->userdata('type') == 3 && !isset($id)): ?>
        <input type="hidden" name="tech_id" value="<?= $_settings-
>userdata('id') ?>">
    <?php endif; ?>
    <?php if($_settings->userdata('type') != 3): ?>
    <fieldset>
        <legend>Assign</legend>
        <div class="row">
            <div class="col-lg-4 col-md-6 col-sm-12 col-xs-12">
```

```
                        <select name="tech_id" id="tech_id" class="form-control
form-control rounded-0">
                            <option value="" disabled <?= !isset($tech_id) ?
"selected" : "" ?>></option>
                            <option value="" <?= isset($tech_id) &&
in_array($tech_id,[null,""]) ? "selected" : "" ?>>Unset</option>
                            <?php
                            $tech_qry = $conn->query("SELECT *,concat(firstname,' ',
lastname) as `name` FROM `users` where `type` = 3 order by `name` asc");
                            while($row = $tech_qry->fetch_array()):
                            ?>
                            <option value="<?= $row['id'] ?>" <?= isset($tech_id)
&& $tech_id == $row['id'] ? "selected" : "" ?><?= $row['name'] ?></option>
                            <?php endwhile; ?>
                        </select>
                    </div>
                </div>
            </fieldset>
            <?php endif; ?>
        </form>
      </div>
    </div>
    <div class="card-footer py-2 text-right">
      <button class="btn btn-primary rounded-0" form="transaction-
form">Save Transaction</button>
      <?php if(!isset($id)): ?>
      <a class="btn btn-default border rounded-0"
href="./?page=transactions">Cancel</a>
      <?php else: ?>
      <a class="btn btn-default border rounded-0"
href="./?page=transactions/view_details&id=<?= $id ?>">Cancel</a>
      <?php endif; ?>
    </div>
  </div>
 </div>
</div>
<noscript id="service-clone">
  <tr>
```

```
        <td class="text-center">
            <button class="btn btn-outline-danger btn-sm rounded-0 rem-service"
type="button"><i class="fa fa-times"></i></button>
        </td>
        <td>
            <input type="hidden" name="service_id[]" value="">
            <input type="hidden" name="service_price[]" value="0">
            <span class="service_name"></span>
        </td>
        <td class="text-right service_price"></td>
    </tr>
</noscript>
<noscript id="product-clone">
    <tr>
        <td class="text-center">
            <button class="btn btn-outline-danger btn-sm rounded-0 rem-product"
type="button"><i class="fa fa-times"></i></button>
        </td>
        <td>
            <input type="hidden" name="product_id[]" value="">
            <input type="hidden" name="product_price[]" value="0">
            <span class="product_name"></span>
        </td>
        <td class=""><input type="number" min="1" class="form-control form-
control-sm rounded-0 text-right" name="product_qty[]" value="1"></td>
        <td class="text-right product_price"></td>
        <td class="text-right product_total"></td>
    </tr>
</noscript>
<script>
    function calc_total_amount(){
        var total = 0;
        $('#service-list tbody tr').each(function(){
            total += parseFloat($(this).find('[name="service_price[]"]').val())
        })
        $('#product-list tbody tr').each(function(){
```

```
        var qty = $(this).find('[name="product_qty[]"]').val()
        qty = qty > 0 ? qty : 0
        total += (parseFloat($(this).find('[name="product_price[]"]').val()) *
parseFloat(qty))
    })
    $('[name="amount"]').val(parseFloat(total))
    $('#amount').text(parseFloat(total).toLocaleString('en-US'))
  }
  function calc_service(){
    var total = 0;
    $('#service-list tbody tr').each(function(){
        total += parseFloat($(this).find('[name="service_price[]"]').val())
    })
    $('#service_total').text(parseFloat(total).toLocaleString('en-US'))
    calc_total_amount()
  }
  function calc_product(){
    var total = 0;

    $('#product-list tbody tr').each(function(){
        var qty = $(this).find('[name="product_qty[]"]').val()
        qty = qty > 0 ? qty : 0
        total += (parseFloat($(this).find('[name="product_price[]"]').val()) *
parseFloat(qty))
    })
    $('#product_total').text(parseFloat(total).toLocaleString('en-US'))
    calc_total_amount()
  }
  $(function(){
    $('select#tech_id').select2({
        placeholder:"Select technician here",
        width:'100%',
        containerCssClass:'form-control form-control-sm rounded-0'
    })
    $('#service_sel').select2({
        placeholder:"Select service here",
```

```
        width:'100%',
        containerCssClass:'form-control form-control-sm rounded-0'
    })
    $('#product_sel').select2({
        placeholder:"Select product here",
        width:'100%',
        containerCssClass:'form-control form-control-sm rounded-0'
    })
    $('#service-list tbody tr').find('.rem-service').click(function(){
        var tr = $(this).closest('tr')
        if(confirm("Are you sure to remove "+(tr.find('.service_name').text())+"
from service list?") === true){
            tr.remove()
            calc_service()
        }
    })
    $('#product-list tbody tr').find('.rem-product').click(function(){
        var tr = $(this).closest('tr')
        if(confirm("Are you sure to remove "+(tr.find('.product_name').text())+"
from product list?") === true){
            tr.remove()
            calc_product()
        }
    })
    $('#product-list tbody tr').find('[name="product_qty[]"]').on('input change',
function(){
        var tr = $(this).closest('tr')
        var qty = $(this).val()
        qty = qty > 0 ? qty : 0
        var total = parseFloat(qty) * parseFloat(price)
        tr.find('.product_total').text(parseFloat(total).toLocaleString())
        calc_product()

    })
    $('#add_service').click(function(){
        if($('#service_sel').val() == null)
```

```
        return false;
        var id = $('#service_sel').val()
        if($('#service-list tbody tr
input[name="service_id[]"][value="'+id+'"]').length > 0){
            alert("Service already on the list.")
            return false;
        }
        var name = $('#service_sel option[value="'+id+'"]').text()
        var price = $('#service_sel option[value="'+id+'"]').attr('data-price')
        var tr = $($('noscript#service-clone').html()).clone()
        tr.find('input[name="service_id[]"]').val(id)
        tr.find('input[name="service_price[]"]').val(price)
        tr.find('.service_name').text(name)
        tr.find('.service_price').text(parseFloat(price).toLocaleString())
        $('#service-list tbody').append(tr)
        calc_service()
        tr.find('.rem-service').click(function(){
            if(confirm("Are you sure to remove "+name+" from service list?") ===
true){
                tr.remove()
                calc_service()
            }
        })
        $('#service_sel').val('').trigger("change")
    })
    $('#add_product').click(function(){
        if($('#product_sel').val() == null)
        return false;
        var id = $('#product_sel').val()
        if($('#product-list tbody tr
input[name="product_id[]"][value="'+id+'"]').length > 0){
            alert("Product already on the list.")
            return false;
        }
        var name = $('#product_sel option[value="'+id+'"]').text()
        var price = $('#product_sel option[value="'+id+'"]').attr('data-price')
```

```
        var tr = $($('noscript#product-clone').html()).clone()
        tr.find('input[name="product_id[]"]').val(id)
        tr.find('input[name="product_price[]"]').val(price)
        tr.find('.product_name').text(name)
        tr.find('.product_price').text(parseFloat(price).toLocaleString())
        tr.find('.product_total').text(parseFloat(price).toLocaleString())
        $('#product-list tbody').append(tr)
        calc_product()
        tr.find('.rem-product').click(function(){
            if(confirm("Are you sure to remove "+name+" from product list?") ===
true){
                tr.remove()
                calc_product()
            }
        })
        tr.find('[name="product_qty[]"]').on('input change', function(){
            var qty = $(this).val()
            qty = qty > 0 ? qty : 0
            var total = parseFloat(qty) * parseFloat(price)
            tr.find('.product_total').text(parseFloat(total).toLocaleString())
            calc_product()

        })
        $('#product_sel').val('').trigger("change")
    })
    $('#product-list, #service-list').find('td, th').addClass('px-2 py-1 align-
middle')
    $('#transaction-form').submit(function(e){
        e.preventDefault();
        var _this = $(this)
         $('.err-msg').remove();
        start_loader();
        $.ajax({
            url:_base_url_+"classes/Master.php?f=save_transaction",
            data: new FormData($(this)[0]),
            cache: false,
```

```
            contentType: false,
            processData: false,
            method: 'POST',
            type: 'POST',
            dataType: 'json',
            error:err=>{
                console.log(err)
                alert_toast("An error occured",'error');
                end_loader();
            },
            success:function(resp){
                if(typeof resp ='object' && resp.status == 'success'){
                    location.href = "./?page=transactions/view_details&id="+resp.tid
                }else if(resp.status == 'failed' && !!resp.msg){
                    var el = $('<div>')
                        el.addClass("alert alert-danger err-msg").text(resp.msg)
                        _this.prepend(el)
                        el.show('slow')
                        $("html, body,.modal").scrollTop(0);
                        end_loader()
                }else{
                    alert_toast("An error occured",'error');
                    end_loader();
                }
            }
        })
    })
})
</script>
```

6. manage_users.php (User Management)

Including a user management file will demonstrate how administrators manage the user accounts and assign roles. It provides a sense of how different roles are managed within the system.

Code:

```
<?php
if(isset($_GET['id'])){
   $user = $conn->query("SELECT * FROM users where id ='{$_GET['id']}' ");
   foreach($user->fetch_array() as $k =>$v){
      $meta[$k] = $v;
   }
}
?>
<?php if($_settings->chk_flashdata('success')): ?>
<script>
   alert_toast("<?php echo $_settings->flashdata('success') ?>",'success')
</script>
<?php endif;?>
<div class="card card-outline rounded-0 card-navy">
   <div class="card-body">
      <div class="container-fluid">
         <div id="msg"></div>
         <form action="" id="manage-user">
            <input type="hidden" name="id" value="<?= isset($meta['id']) ?
$meta['id'] : '' ?>">
            <div class="form-group">
               <label for="name">First Name</label>
               <input type="text" name="firstname" id="firstname" class="form-
control" value="<?php echo isset($meta['firstname']) ? $meta['firstname']: '' ?>"
required>
            </div>
            <div class="form-group">
               <label for="name">Last Name</label>
               <input type="text" name="lastname" id="lastname" class="form-
control" value="<?php echo isset($meta['lastname']) ? $meta['lastname']: '' ?>"
required>
            </div>
```

```
            <div class="form-group">
                <label for="username">Username</label>
                <input type="text" name="username" id="username" class="form-
control" value="<?php echo isset($meta['username']) ? $meta['username']: '' ?>"
required  autocomplete="off">
            </div>
            <div class="form-group">
                <label for="password"><?= isset($meta['id']) ? "New" : "" ?>
Password</label>
                <input type="password" name="password" id="password"
class="form-control" value="" autocomplete="off">
                <?php if(isset($meta['id'])): ?>
                <small><i>Leave this blank if you dont want to change the
password.</i></small>
                <?php endif; ?>
            </div>
            <div class="form-group">
                <label for="type" class="control-label">Type</label>
                <select name="type" id="type" class="form-control form-control-
sm rounded-0" required>
                <option value="1" <?php echo isset($meta['type']) &&
$meta['type'] == 1 ? 'selected' : '' ?>>Administrator</option>
                <option value="2" <?php echo isset($meta['type']) &&
$meta['type'] == 2 ? 'selected' : '' ?>>Staff</option>
                <option value="3" <?php echo isset($meta['type']) &&
$meta['type'] == 3 ? 'selected' : '' ?>>Technician</option>
                </select>
            </div>
            <div class="form-group">
                <label for="" class="control-label">Avatar</label>
                <div class="custom-file">
                  <input type="file" class="custom-file-input rounded-circle"
id="customFile" name="img" onchange="displayImg(this,$(this))"
accept="image/png, image/jpeg">
                  <label class="custom-file-label" for="customFile">Choose
file</label>
                </div>
            </div>
            <div class="form-group d-flex justify-content-center">
```

```
                <img src="<?php echo validate_image(isset($meta['avatar']) ?
$meta['avatar'] :'') ?>" alt="" id="cimg" class="img-fluid img-thumbnail">
            </div>
        </form>
    </div>
  </div>
  <div class="card-footer">
        <div class="col-md-12">
            <div class="row">
                <button class="btn btn-sm btn-primary rounded-0 mr-3"
form="manage-user">Save User Details</button>
                <a href="./?page=user/list" class="btn btn-sm btn-default border
rounded-0" form="manage-user"><i class="fa fa-angle-left"></i> Cancel</a>
            </div>
        </div>
    </div>
</div>
<style>
  img#cimg{
    height: 15vh;
    width: 15vh;
    object-fit: cover;
    border-radius: 100% 100%;
  }
</style>
<script>
  function displayImg(input,_this) {
    if (input.files && input.files[0]) {
        var reader = new FileReader();
        reader.onload = function (e) {
            $('#cimg').attr('src', e.target.result);
        }

        reader.readAsDataURL(input.files[0]);
    }else{
        $('#cimg').attr('src', "<?php echo validate_image(isset($meta['avatar']) ?
$meta['avatar'] :'') ?>");
```

```
        }
    }
    $('#manage-user').submit(function(e){
        e.preventDefault();
        start_loader()
        $.ajax({
            url:_base_url_+'classes/Users.php?f=save',
            data: new FormData($(this)[0]),
            cache: false,
            contentType: false,
            processData: false,
            method: 'POST',
            type: 'POST',
            success:function(resp){
                if(resp ==1){
                    location.href='./?page=user/list'
                }else{
                    $('#msg').html('<div class="alert alert-danger">Username already
exist</div>')
                    end_loader()
                }
            }
        })
    })

</script>
```

7. view_services.php or view_product.php (Public-side functionality)

A file that handles the public view of services or products will give insight into how the public-facing portion of the site functions, including listing services/products.

Code (view_services.php):

```
<?php

require_once('../../config.php');
if(isset($_GET['id']) && $_GET['id'] > 0){
   $qry = $conn->query("SELECT * from `service_list` where id = '{$_GET['id']}' ");
   if($qry->num_rows > 0){
      foreach($qry->fetch_assoc() as $k => $v){
         $$k=$v;
      }
   }
}
?>
<style>
   #uni_modal .modal-footer{
      display:none;
   }
</style>
<div class="container-fluid">
   <dl>
      <dt class="text-muted">Service</dt>
      <dd class="pl-4"><?= isset($name) ? $name : "" ?></dd>
      <dt class="text-muted">Description</dt>
      <dd class="pl-4"><?= isset($description) ? $description : '' ?></dd>
      <dt class="text-muted">Price</dt>
      <dd class="pl-4"><?= isset($price) ? format_num($price) : '' ?></dd>
      <dt class="text-muted">Status</dt>
      <dd class="pl-4">
         <?php if($status == 1): ?>
            <span class="badge badge-success px-3 rounded-pill">Active</span>
         <?php else: ?>
```

```
            <span class="badge badge-danger px-3 rounded-
pill">Inactive</span>
        <?php endif; ?>
      </dd>
   </dl>
   <div class="clear-fix my-3"></div>
   <div class="text-right">
      <button class="btn btn-sm btn-dark bg-gradient-dark btn-flat"
type="button" data-dismiss="modal"><i class="fa fa-times"></i>
Close</button>
   </div>
</div>
```

Code (view_products.php):

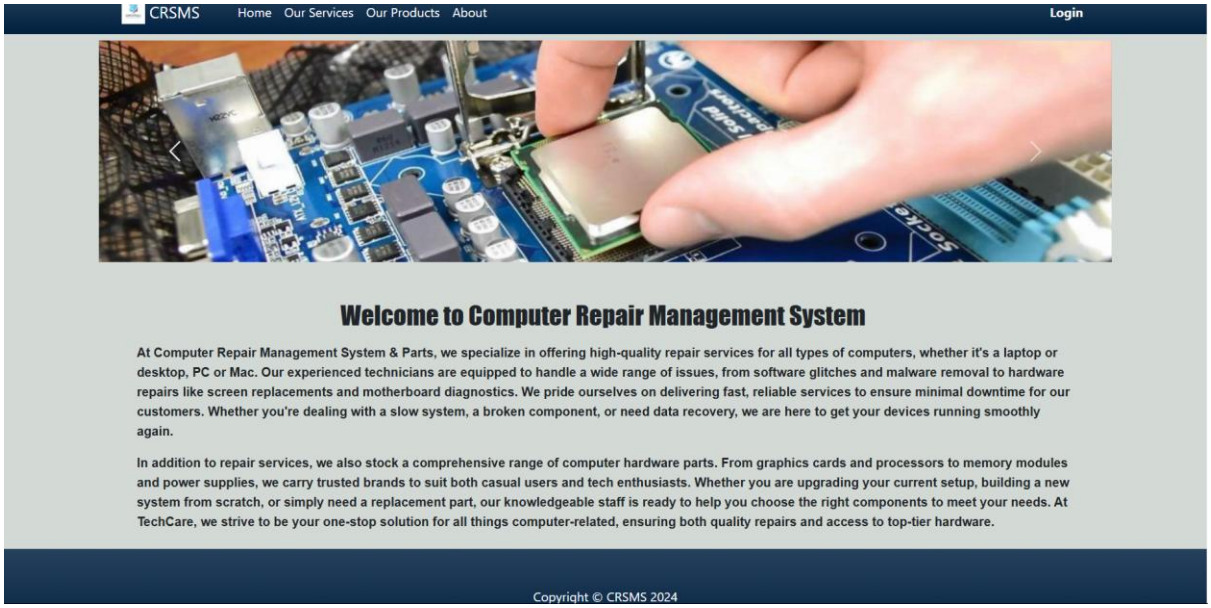
```
<?php

require_once('./../config.php');
if(isset($_GET['id']) && $_GET['id'] > 0){
   $qry = $conn->query("SELECT * from `product_list` where id = '{$_GET['id']}'
");
   if($qry->num_rows > 0){
      foreach($qry->fetch_assoc() as $k => $v){
         $$k=$v;
      }
   }
}
?>
<style>
   #uni_modal .modal-footer{
      display:none;
   }
   #cimg{
      width:100%;
      max-height:20vh;
      object-fit:scale-down;
```

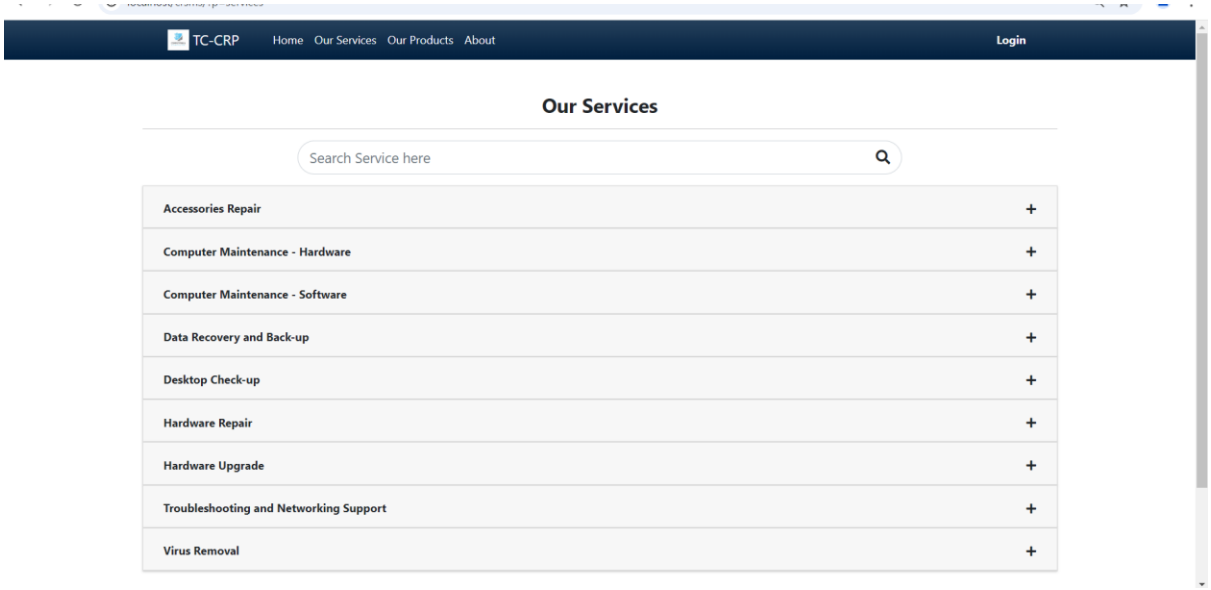
```
        object-position:center center
    }
</style>
<div class="container-fluid">
    <div class="text-center">
            <img src="<?= validate_image(isset($image_path) ? $image_path : '') ?>"
alt="" id="cimg" class="img-fluid bg-gradient-dark w-100">
    </div>
    <dl>
        <dt class="text-muted">Product</dt>
        <dd class="pl-4"><?= isset($name) ? $name : "" ?></dd>
        <dt class="text-muted">Description</dt>
        <dd class="pl-4"><?= isset($description) ? $description : '' ?></dd>
        <dt class="text-muted">Price</dt>
        <dd class="pl-4"><?= isset($price) ? format_num($price) : '' ?></dd>
    </dl>
    <div class="clear-fix my-3"></div>
    <div class="text-right">
        <button class="btn btn-sm btn-dark bg-gradient-dark btn-flat"
type="button" data-dismiss="modal"><i class="fa fa-times"></i>
Close</button>
    </div>
</div>
```

Output:

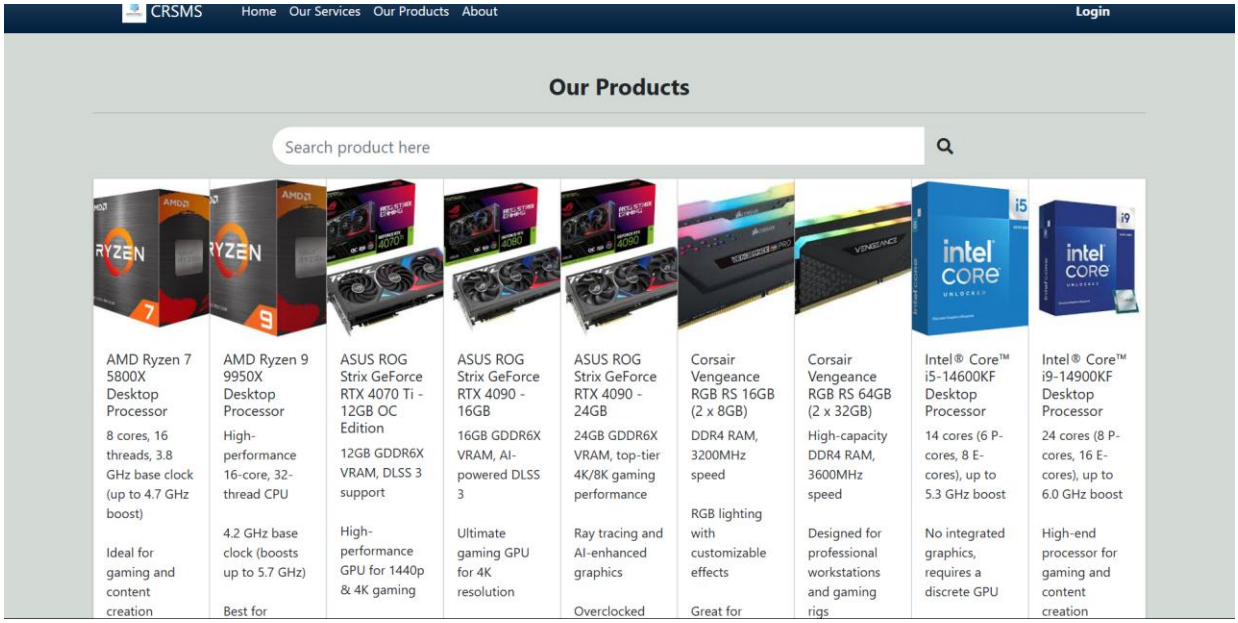
Home Page



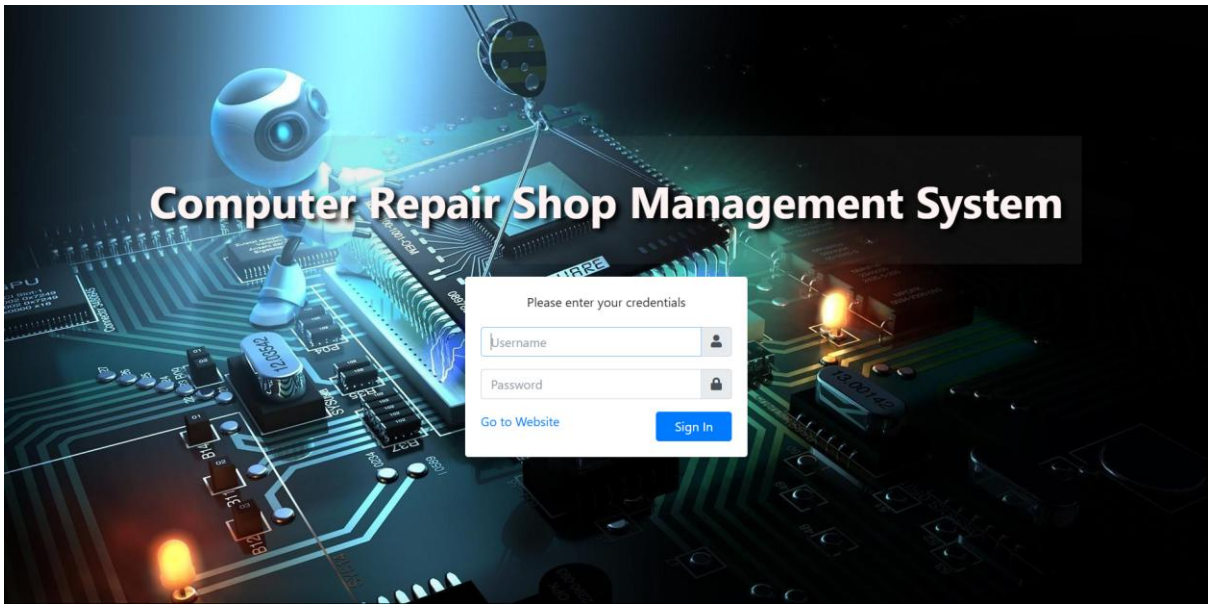
Service Offered Page (Public-Site)



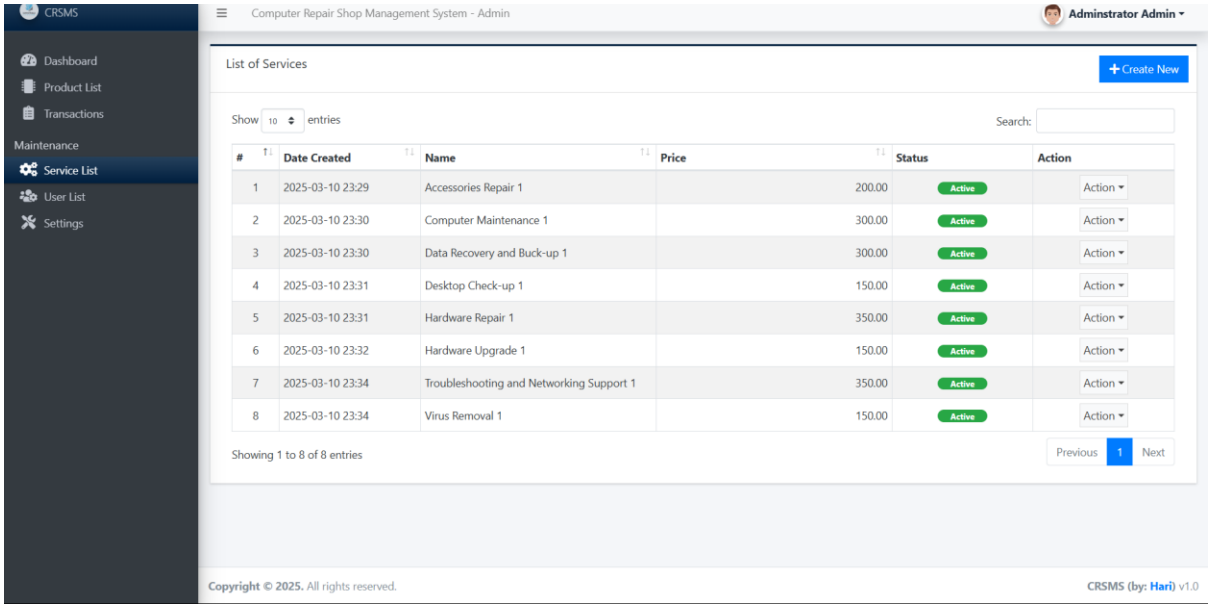
Product Offered Page (Public-Site)



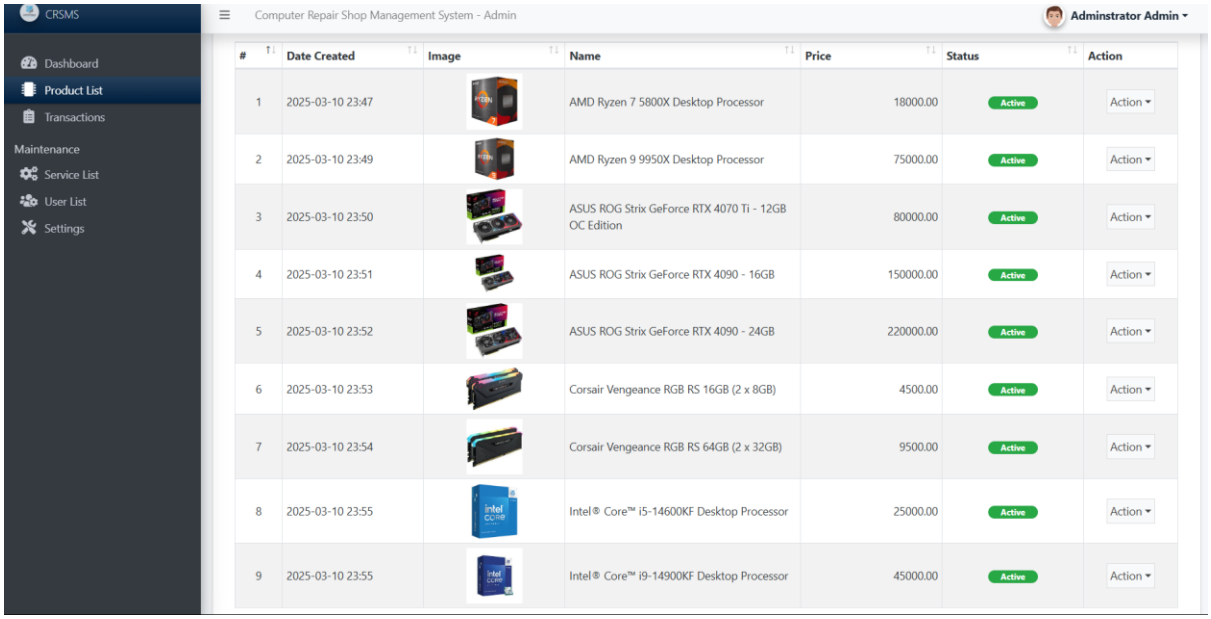
Login Page



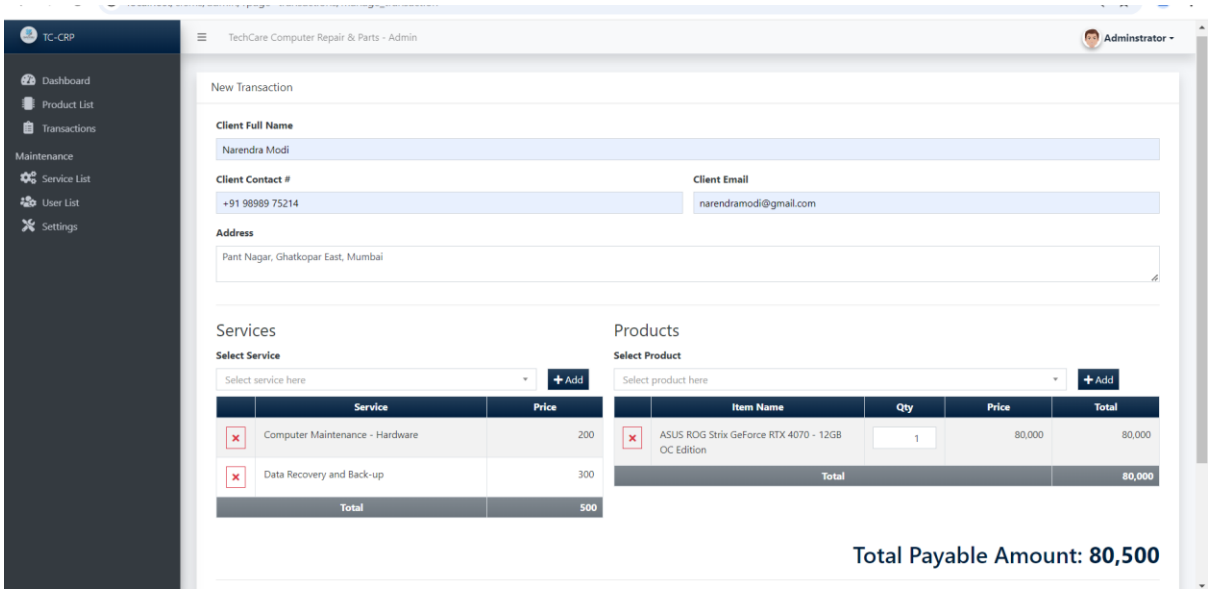
Service List (Administrator-Side)



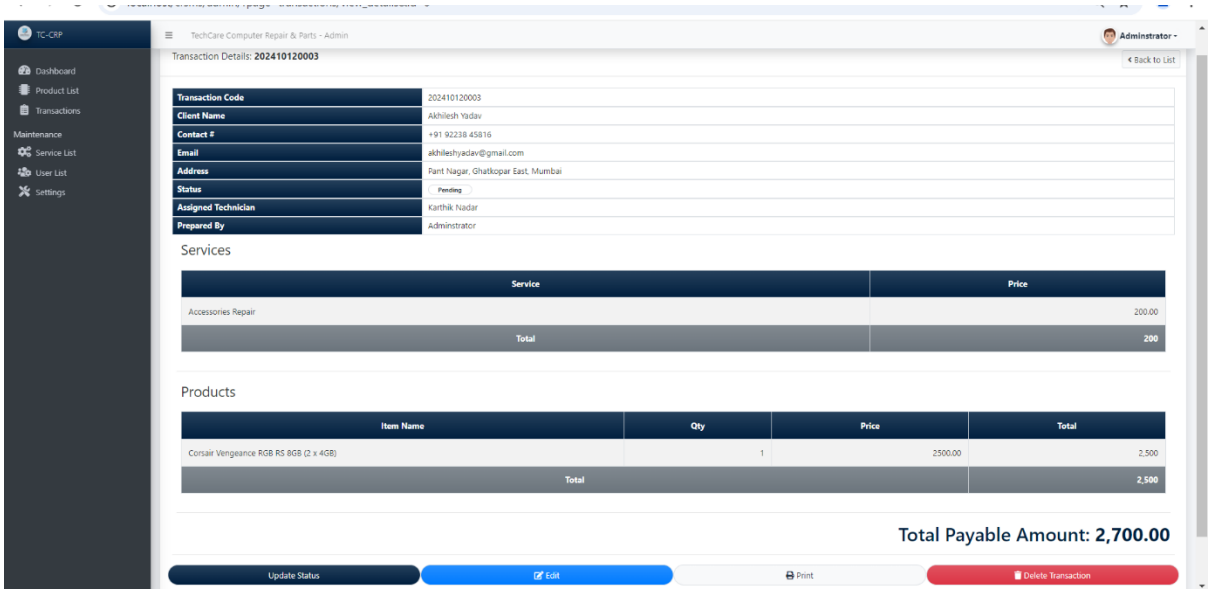
Product List (Admin/Staff-Side)



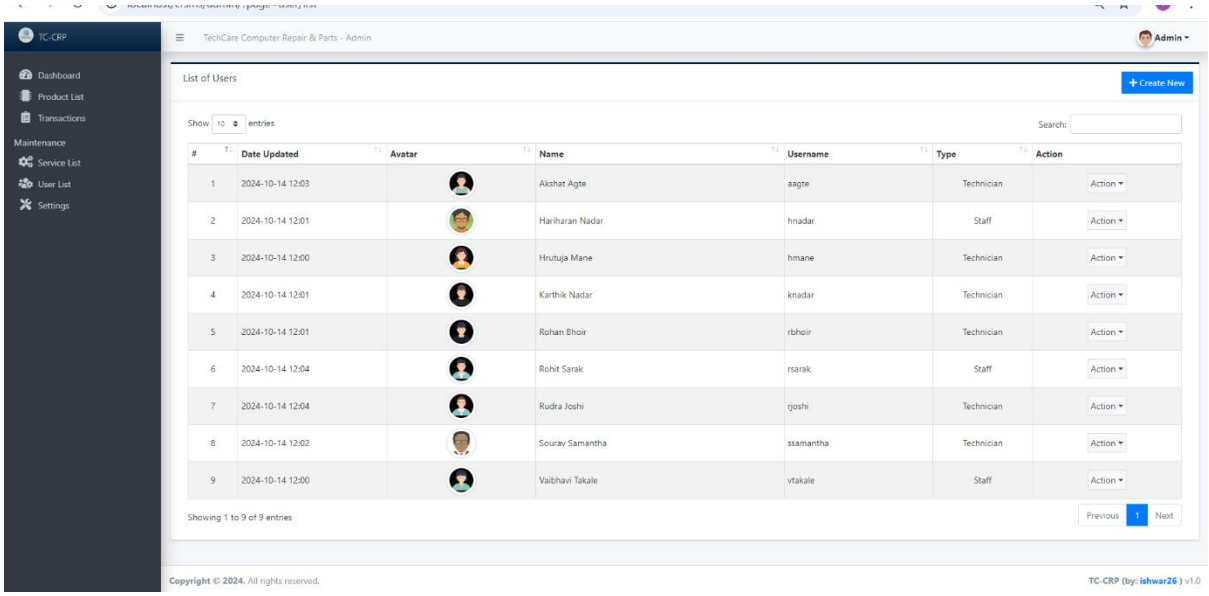
Transaction List



Transaction Details Page



User List (Admin-Side)



Conclusion:

In conclusion, the coding section of the Computer Repair Shop Management System showcases the foundational code that drives the application's functionality and user experience. Each included file has been selected to illustrate the core components of the system, from establishing database connections to managing user roles and handling transactions. The structured approach taken in the development process not only enhances the system's reliability but also provides a scalable framework that can be easily adapted to future needs. By understanding the logic and flow of the code presented, readers can appreciate the intricacies involved in creating a robust web application that effectively addresses the operational requirements of a computer repair shop. This section serves as a vital resource for developers looking to gain insights into best practices in PHP development and database management, while also providing a foundation for further exploration and enhancement of the system.

FUTURE ENHANCEMENT

While the CRSMS has achieved significant milestones, there is always room for improvement. To maintain its relevance and effectiveness, future iterations of the system should focus on:

1. **Enhanced Features**: Incorporating advanced search and filtering functionalities will improve user experience, making it easier for users to locate specific services and products. Additionally, expanding reporting capabilities to provide detailed analytics on transactions and service performance will empower users to make data-driven decisions.
2. **User Training and Support**: Establishing a robust training program will help onboard new users effectively and ensure that existing users can take full advantage of the system's features. Providing comprehensive user manuals, online tutorials, and a feedback mechanism will create a supportive ecosystem for continuous improvement.
3. **Scalability and Performance Monitoring**: As the business grows, the system must be able to scale accordingly. Implementing monitoring tools to assess system performance and user load will ensure that the CRSMS continues to function optimally, even as more users and transactions are added.
4. **Integration with Other Systems**: Exploring potential integrations with third-party tools or platforms, such as accounting software or customer relationship management (CRM) systems, could enhance the functionality of the CRSMS. This would provide users with a more comprehensive solution for managing all aspects of their business.

In conclusion, the Computer Repair Shop Management System represents a significant advancement for computer repair businesses, combining automation, user-friendly design, and effective management tools into a single platform. The successful implementation of the system has demonstrated its potential to improve operational efficiency, enhance customer service, and facilitate growth in a competitive market.

REFERENCES

Play Store Apps and websites:

1. **RepairDesk POS**. (2024). RepairDesk Inc. *Google Play Store*. Retrieved from https://play.google.com/store/apps/details?id=com.repairdesk.pos

- This app provides **repair job tracking, inventory management, invoicing, and CRM**—features that align closely with your CRSMS project.

2. **RepairShopr**. (2024). *Computer Repair Shop Management Software*. Retrieved from https://www.repairshopr.com

- A **web-based solution** offering **ticketing, repair tracking, customer management, and financial reporting**, making it highly relevant to your project.

YouTube Videos:

1. Advanced Java Tutorials. (2024, March 1). CodeWithHarry. YouTube.
2. Web Development Basics. (2024, February 15). FreeCodeCamp. YouTube.

Software & Tools:

- XAMPP. (2024). Apache Friends.
- MySQL. (2024). Oracle Corporation.
- Bootstrap. (2024). Bootstrap Team.
- GitHub. (2024). Microsoft.

PROJECT LINK

Link:
https://drive.google.com/drive/folders/1aW_wDzcSLWlcSEiUs1iJjqpWPpfifPnn?usp=sharing